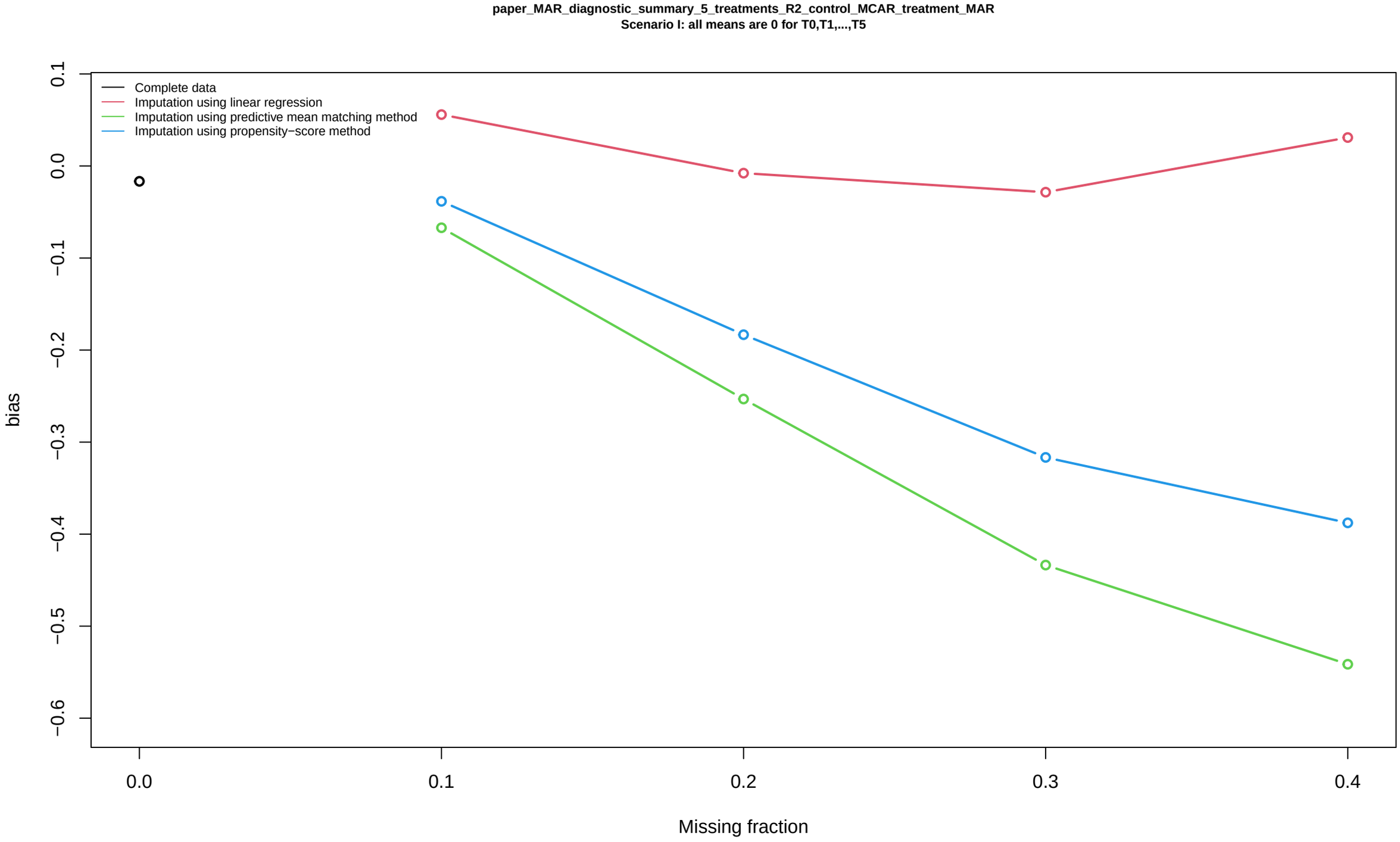
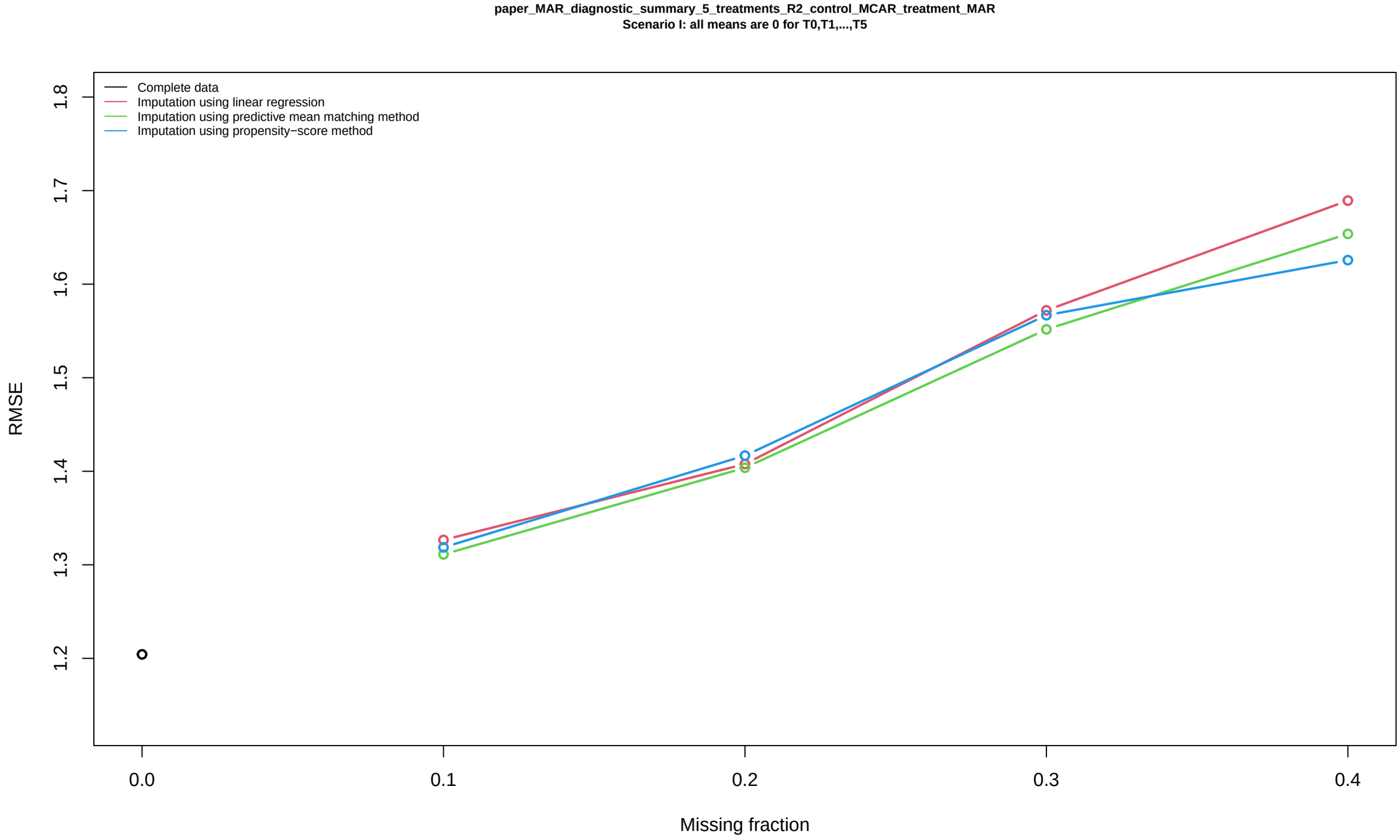
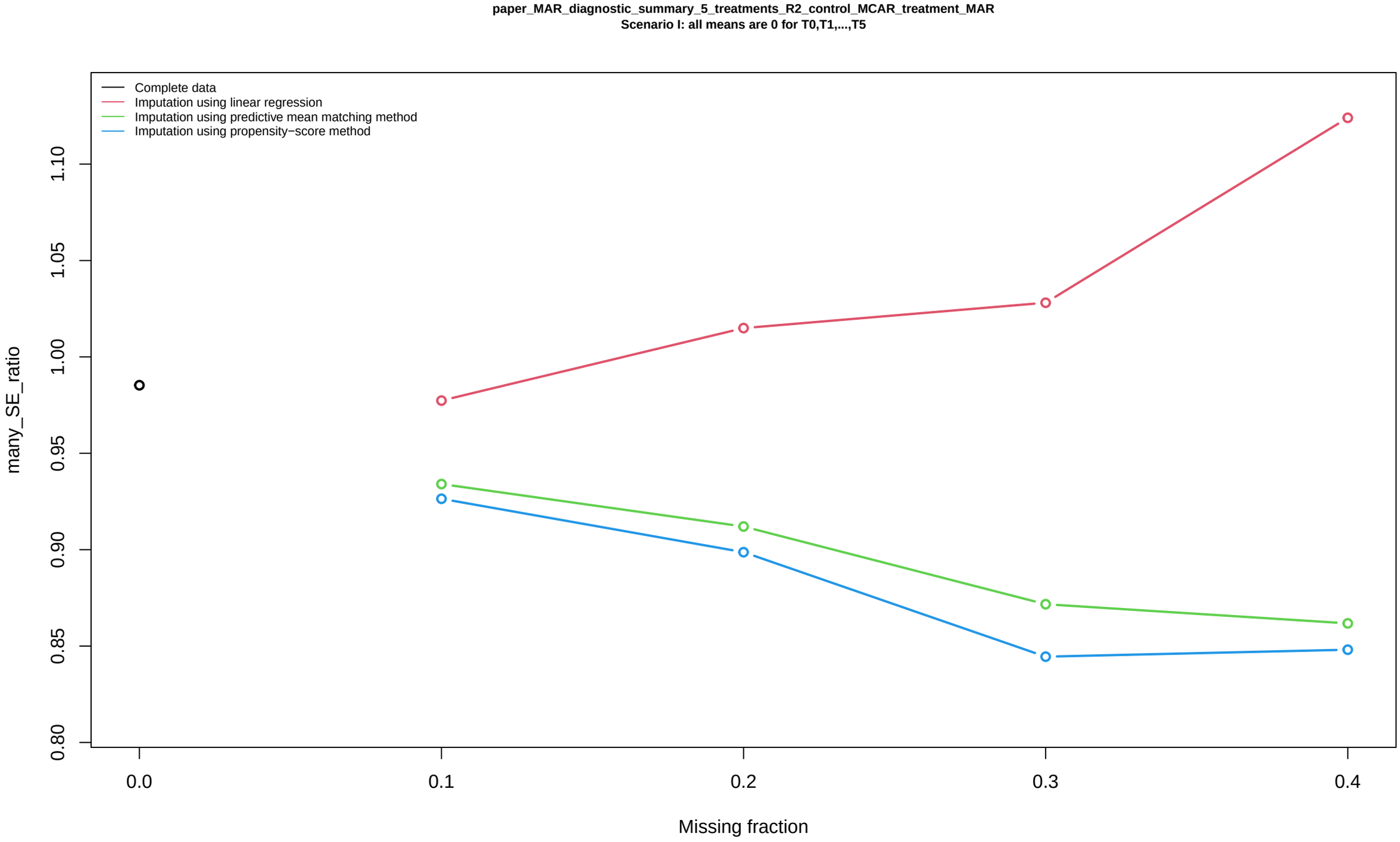
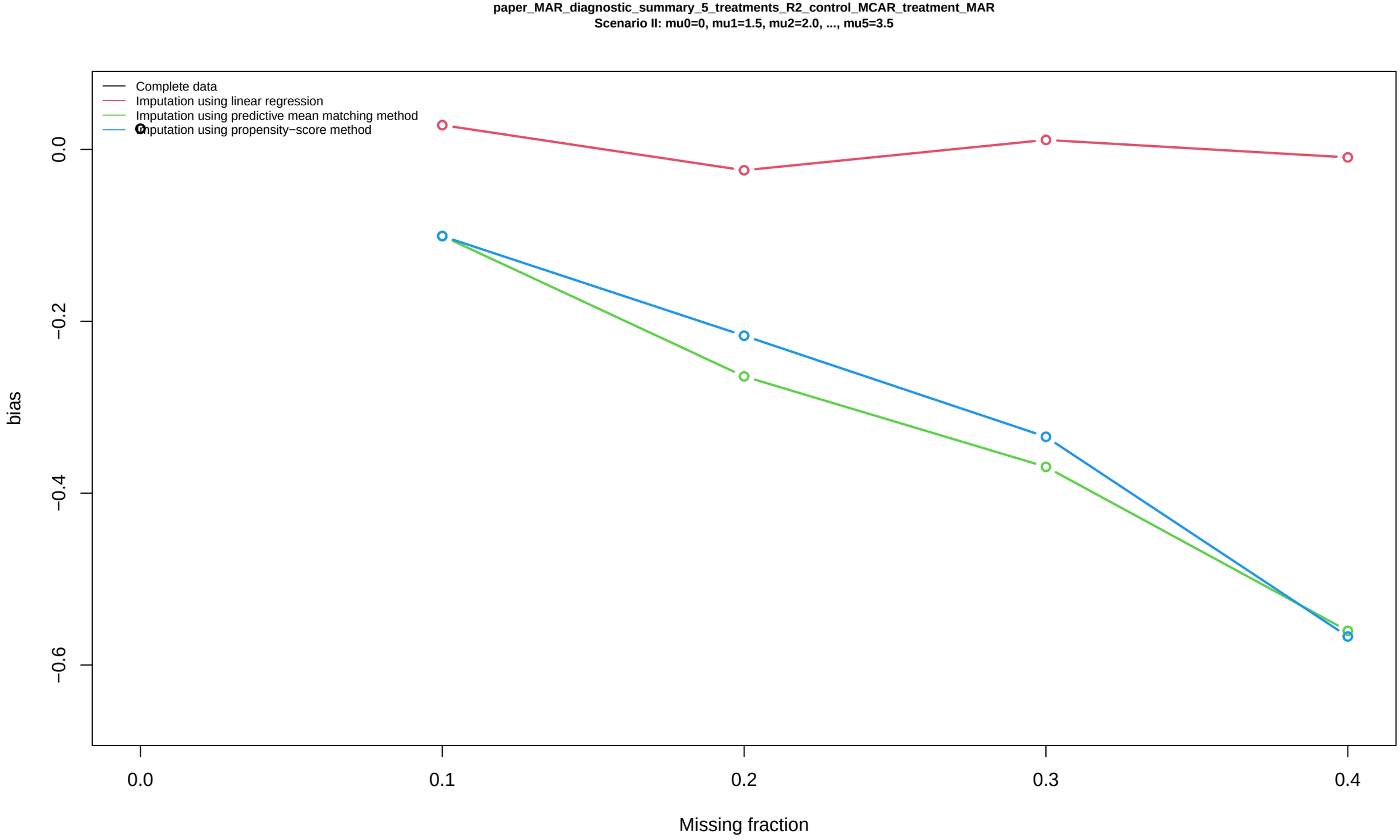
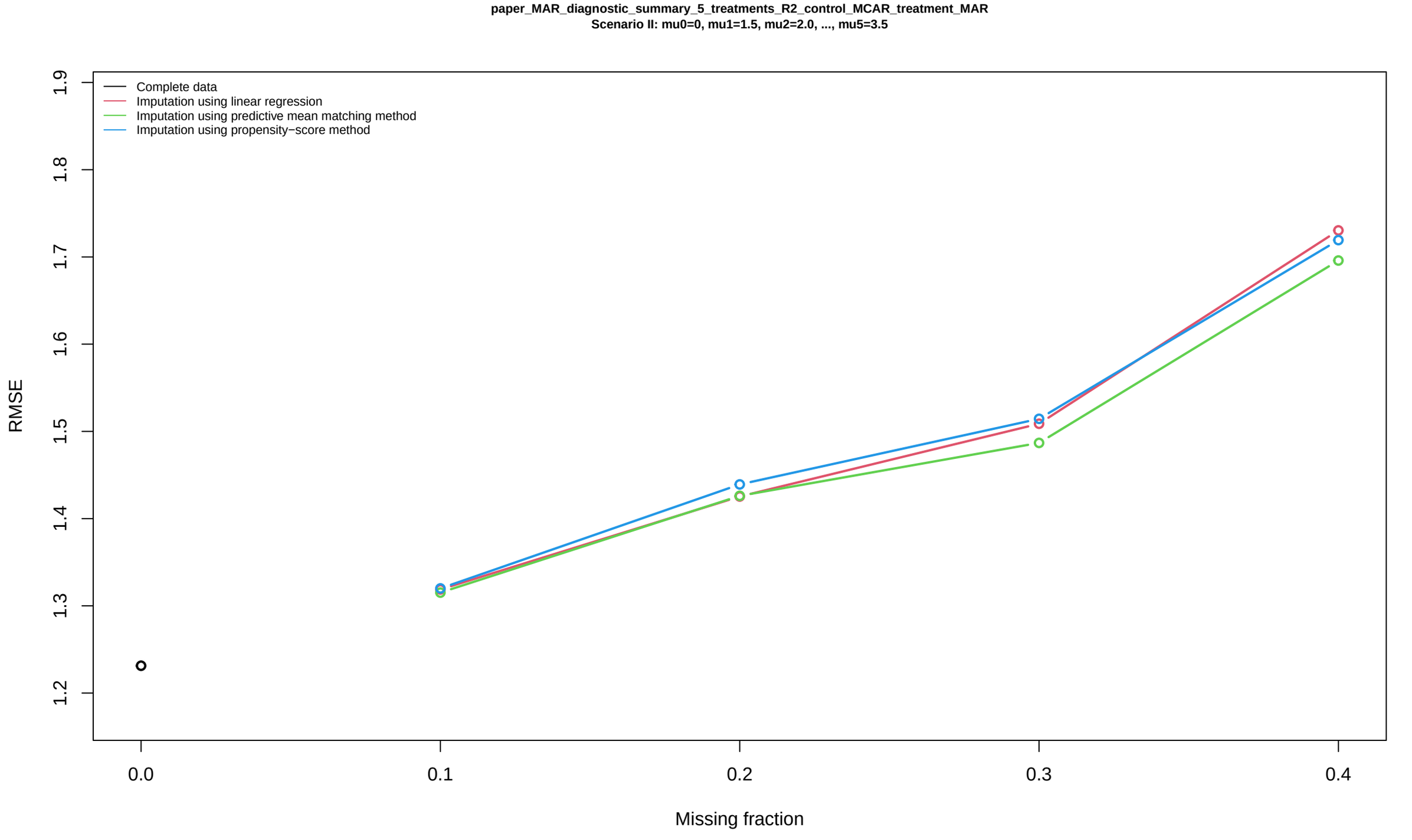
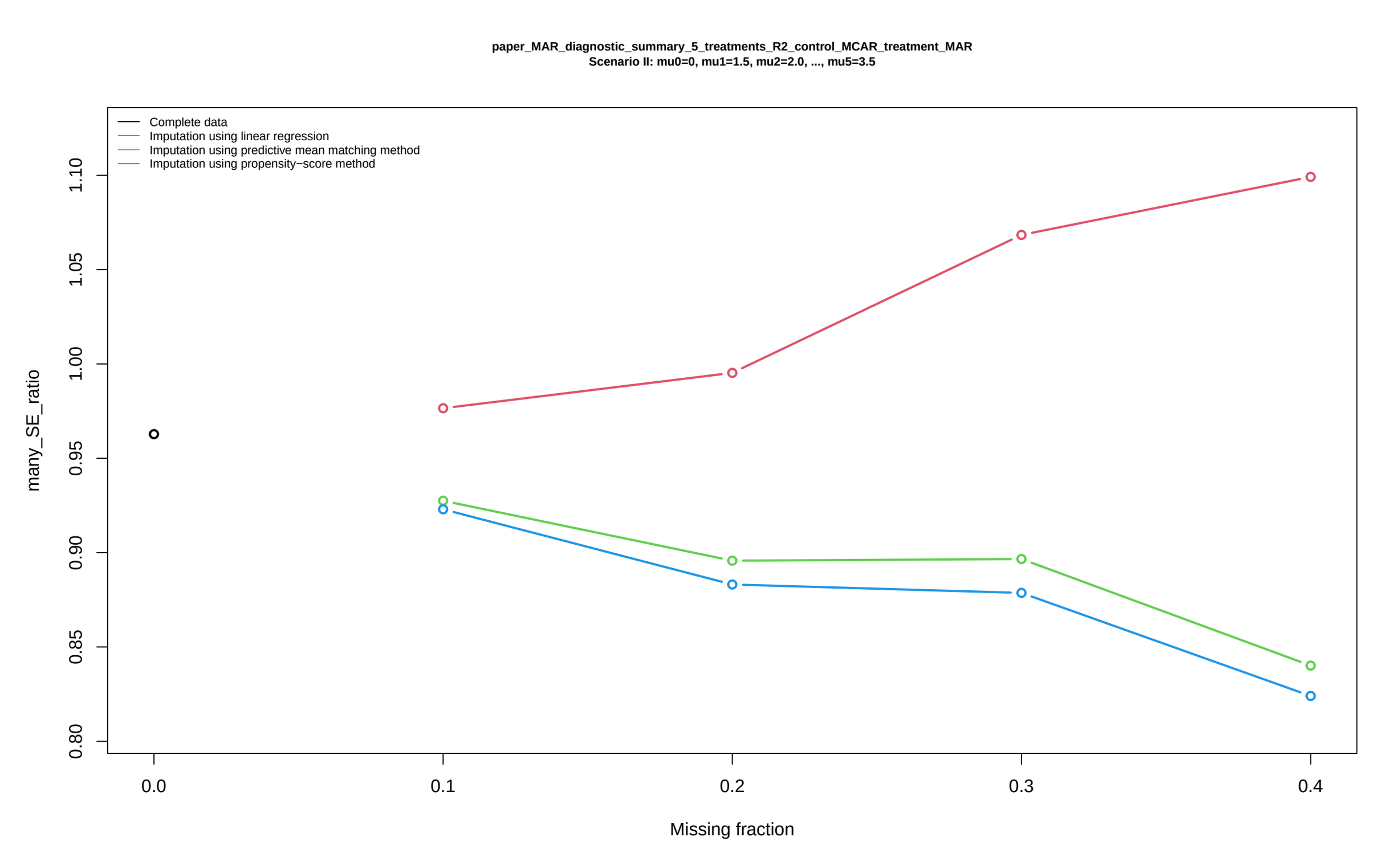

paper_MAR_diagnostic_summary_5_treatments_R2_control_MCAR_treatment_MAR
Scenario I: all means are 0 for T0,T1,...,T5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
bias
0.1
0.0
−0.1
−0.2
−0.3
−0.4
−0.5
−0.6
0.0
0.1
0.2
0.3
0.4
Missing fraction

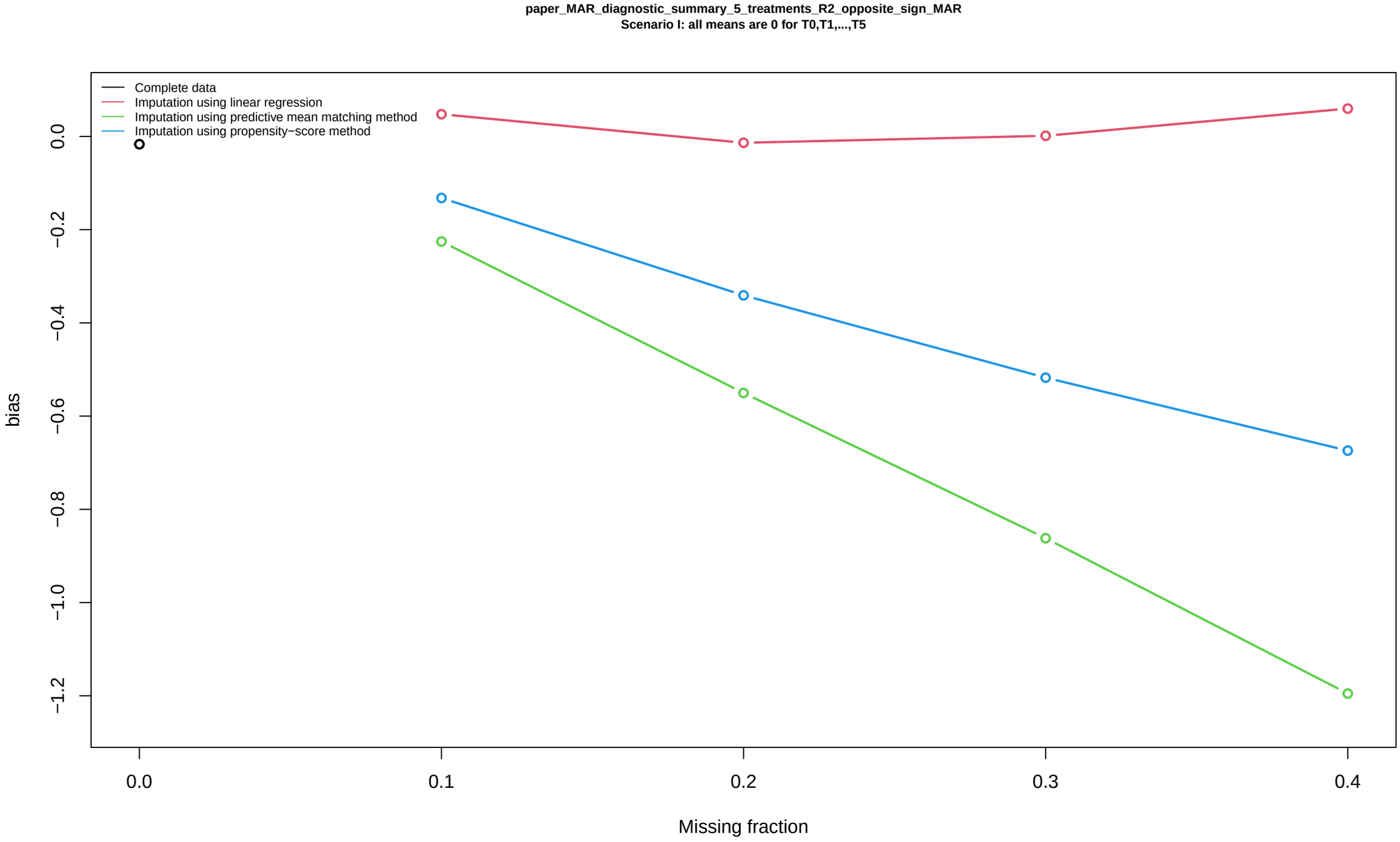

paper_MAR_diagnostic_summary_5_treatments_R2_control_MCAR_treatment_MAR
Scenario I: all means are 0 for T0,T1,...,T5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
RMSE
1.2
1.3
1.4
1.5
1.6
1.7
1.8
0.0
0.1
0.2
0.3
0.4
Missing fraction

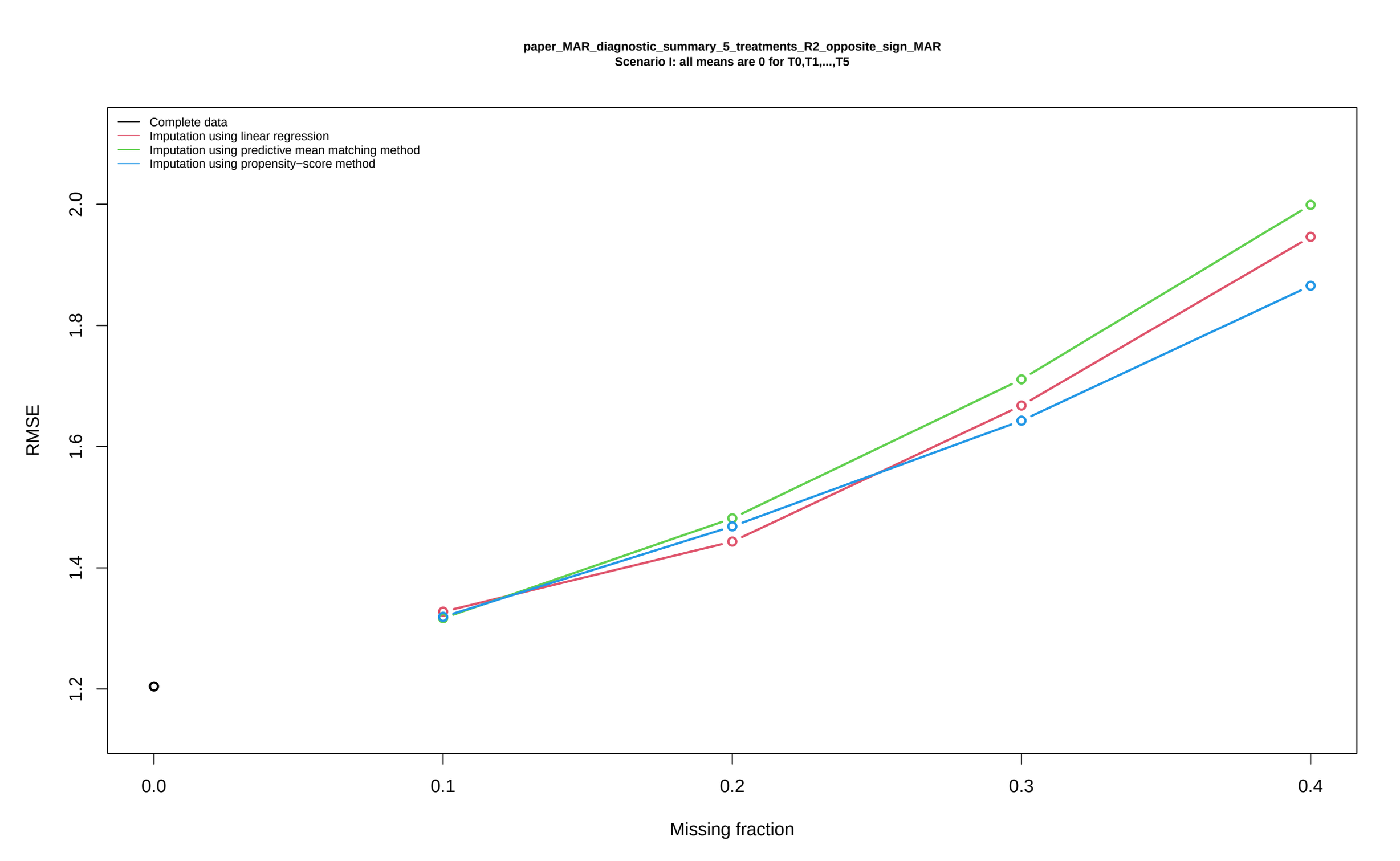

paper_MAR_diagnostic_summary_5_treatments_R2_control_MCAR_treatment_MAR
Scenario I: all means are 0 for T0,T1,...,T5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
many_SE_ratio
0.80
0.85
0.90
0.95
1.00
1.05
1.10
0.0
0.1
0.2
0.3
0.4
Missing fraction

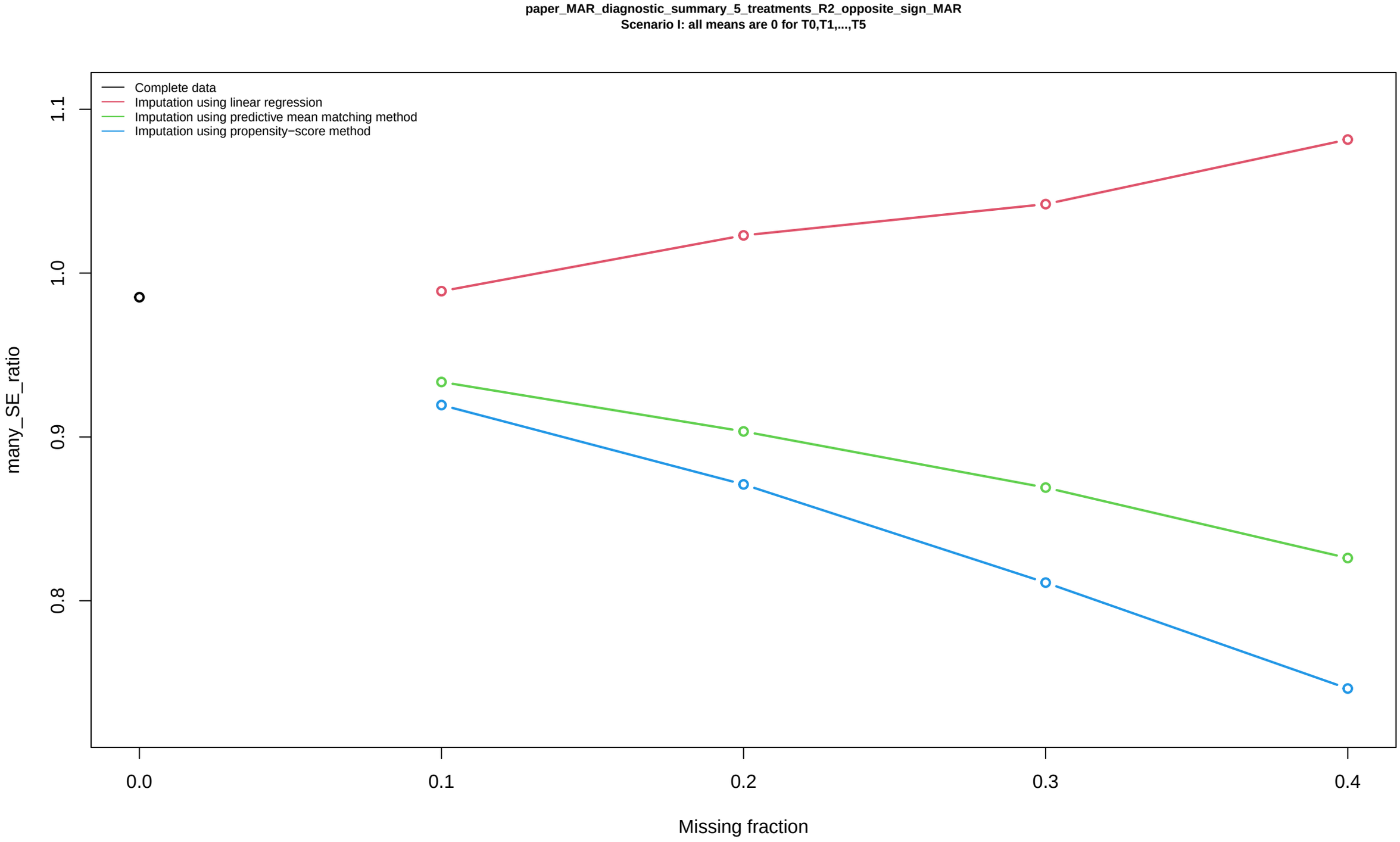

paper_MAR_diagnostic_summary_5_treatments_R2_control_MCAR_treatment_MAR
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
bias
0.0
−0.2
−0.4
−0.6
0.0
0.1
0.2
0.3
0.4
Missing fraction

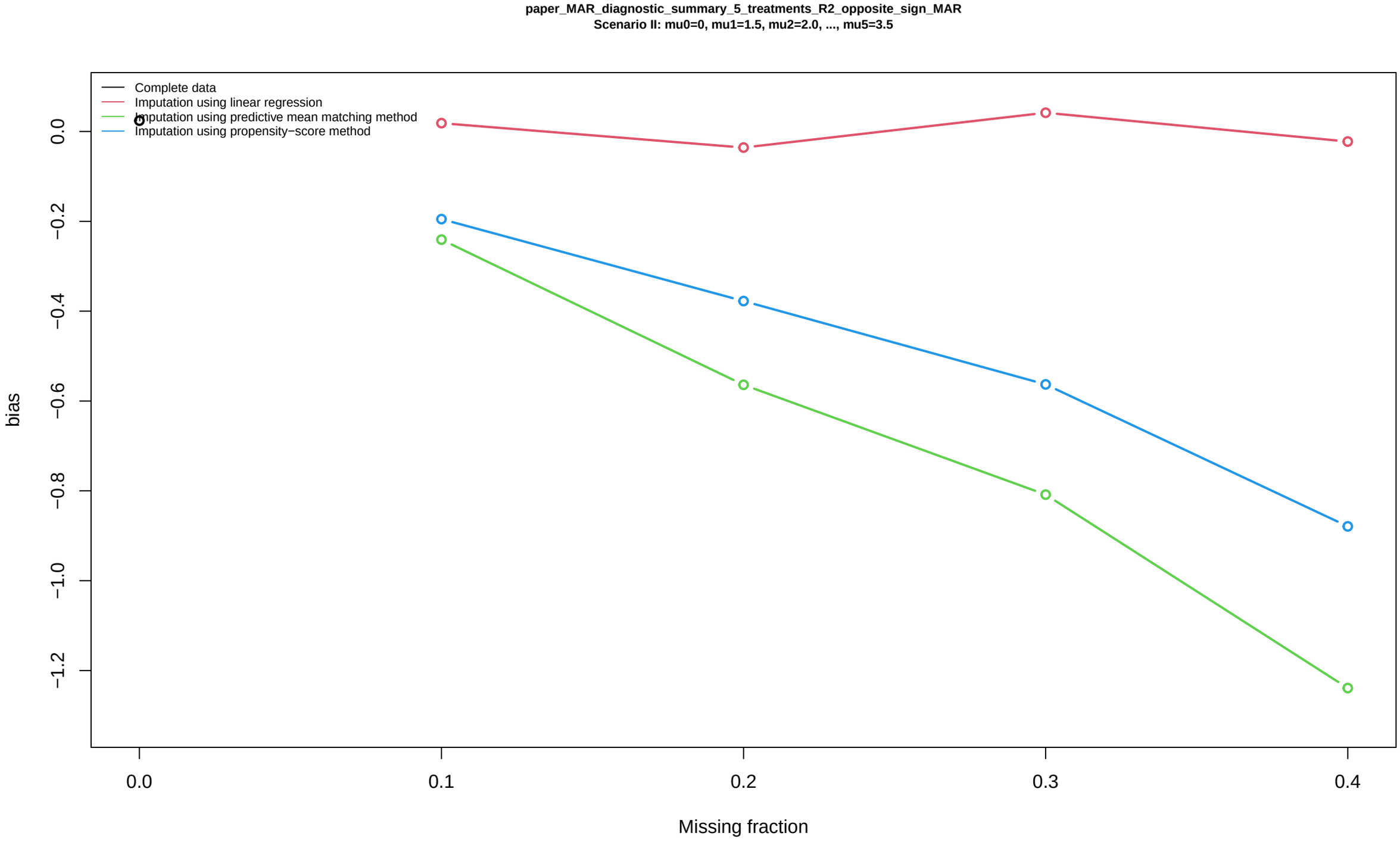

paper_MAR_diagnostic_summary_5_treatments_R2_control_MCAR_treatment_MAR
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
RMSE
1.2
1.3
1.4
1.5
1.6
1.7
1.8
1.9
0.0
0.1
0.2
0.3
0.4
Missing fraction

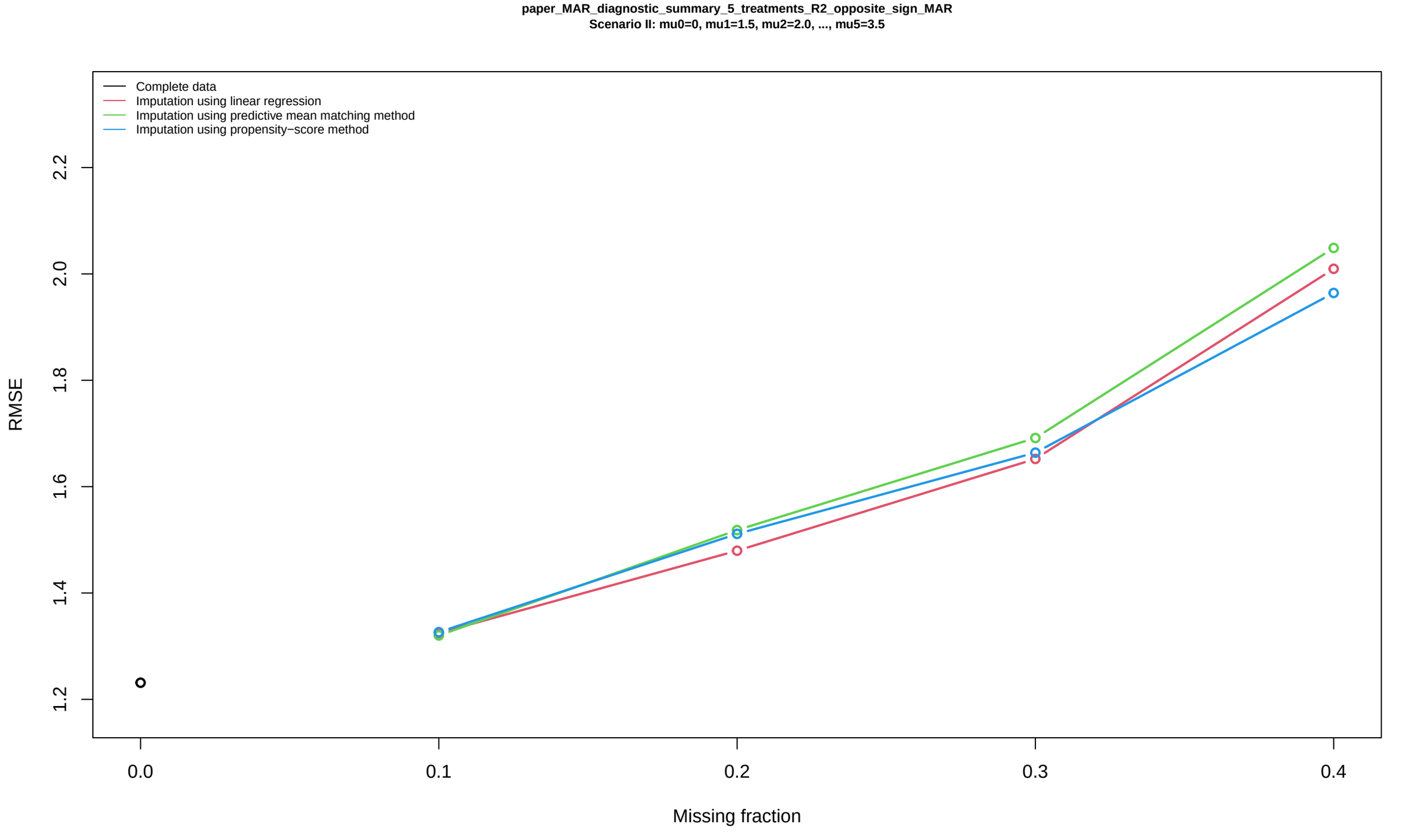

paper_MAR_diagnostic_summary_5_treatments_R2_control_MCAR_treatment_MAR
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
many_SE_ratio
0.80
0.85
0.90
0.95
1.00
1.05
1.10
0.0
0.1
0.2
0.3
0.4
Missing fraction

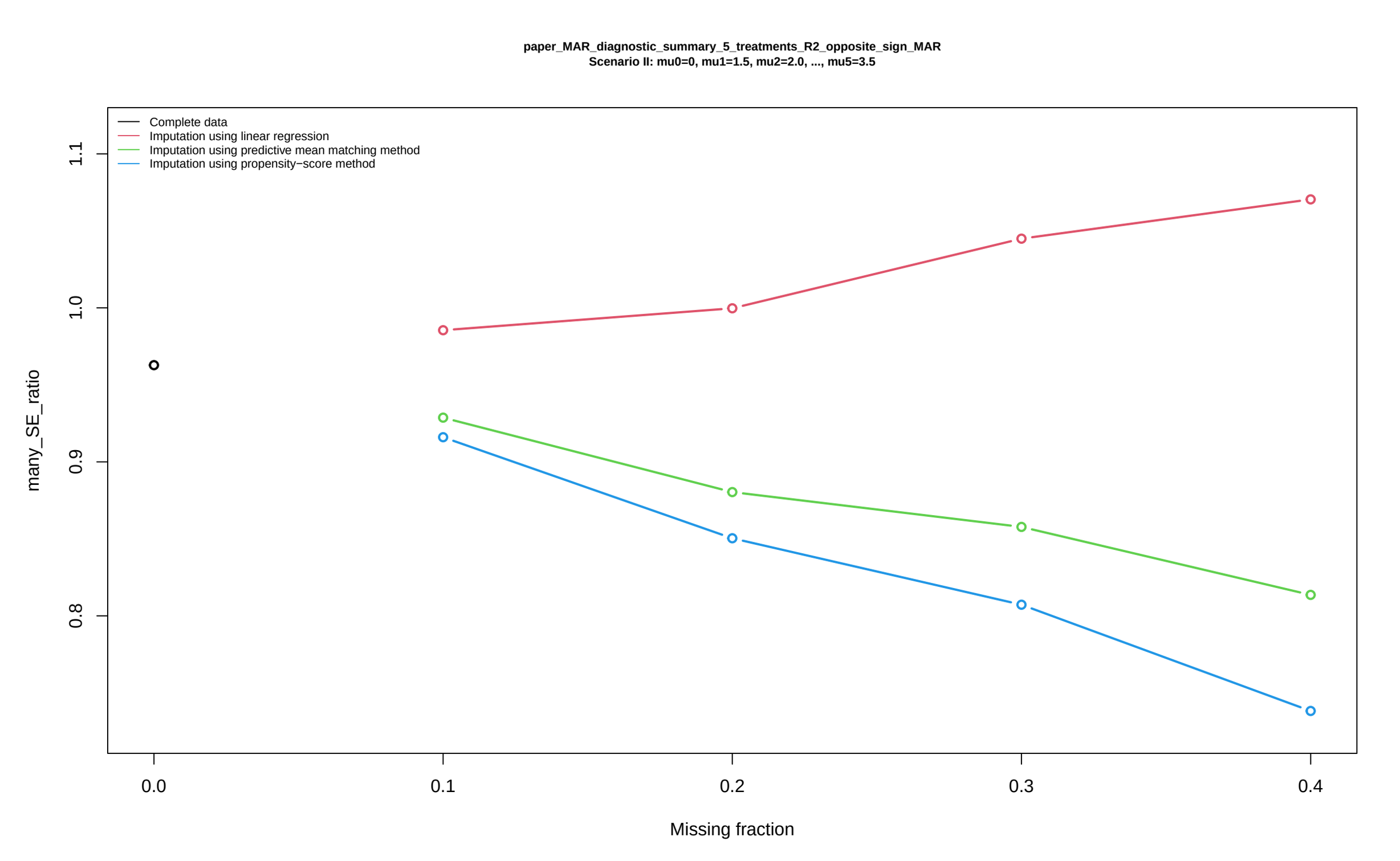

paper_MAR_diagnostic_summary_5_treatments_R2_opposite_sign_MAR
Scenario I: all means are 0 for T0,T1,...,T5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
bias
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.2
0.0
0.1
0.2
0.3
0.4
Missing fraction

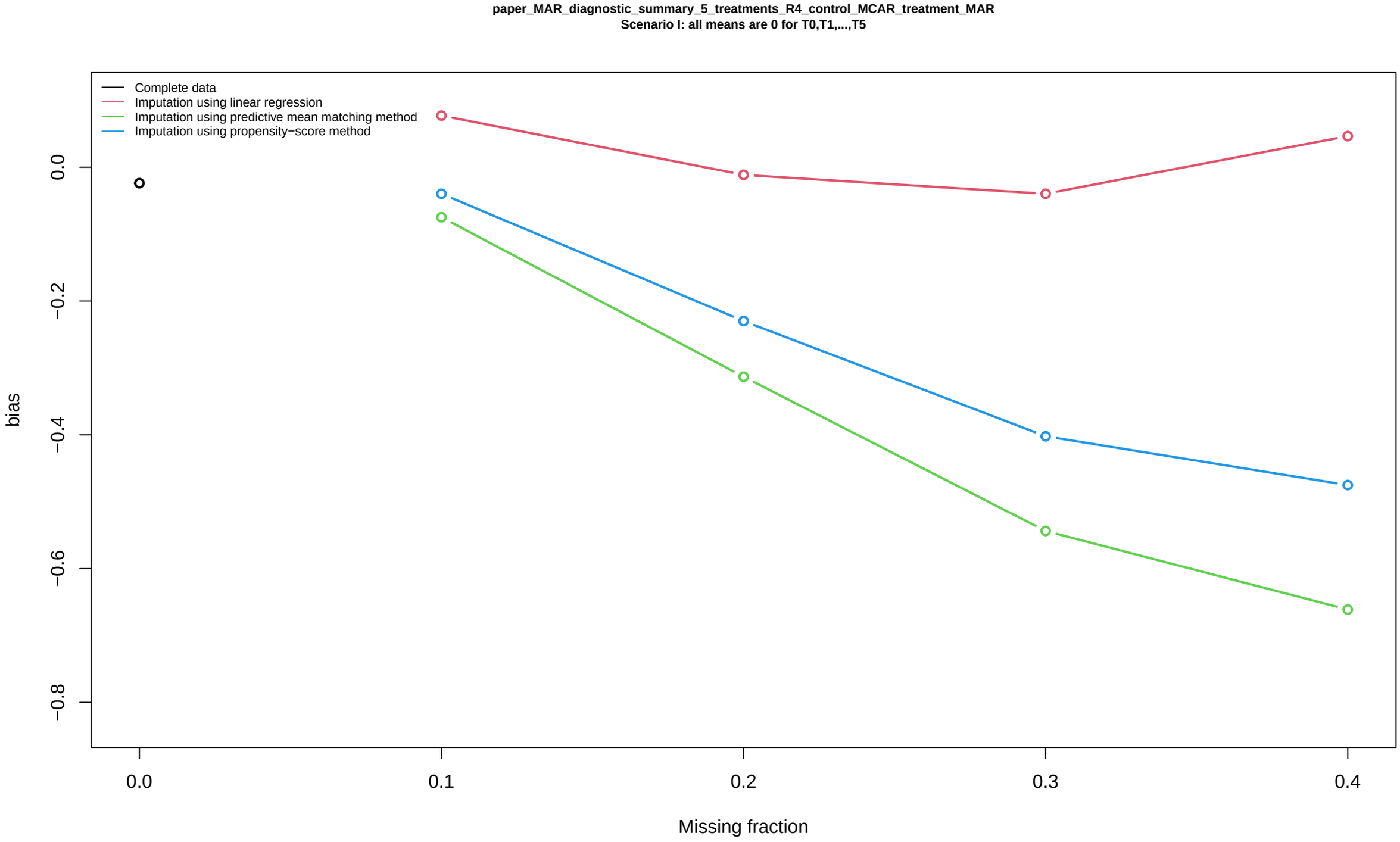

paper_MAR_diagnostic_summary_5_treatments_R2_opposite_sign_MAR
Scenario I: all means are 0 for T0,T1,...,T5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
RMSE
2.0
1.8
1.6
1.4
1.2
0.0
0.1
0.2
0.3
0.4
Missing fraction

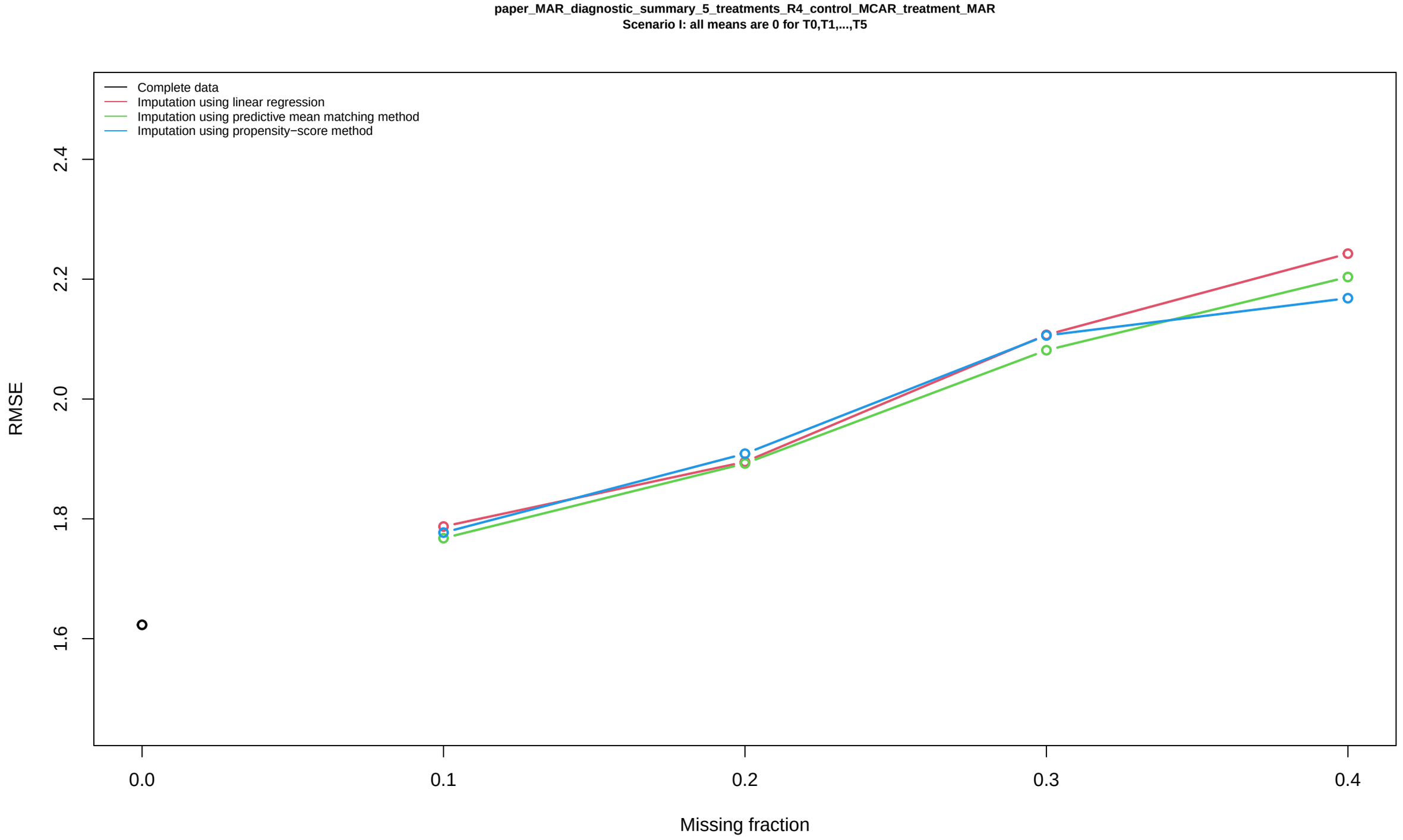

paper_MAR_diagnostic_summary_5_treatments_R2_opposite_sign_MAR
Scenario I: all means are 0 for T0,T1,...,T5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
many_SE_ratio
1.1
1.0
0.9
0.8
0.0
0.1
0.2
0.3
0.4
Missing fraction

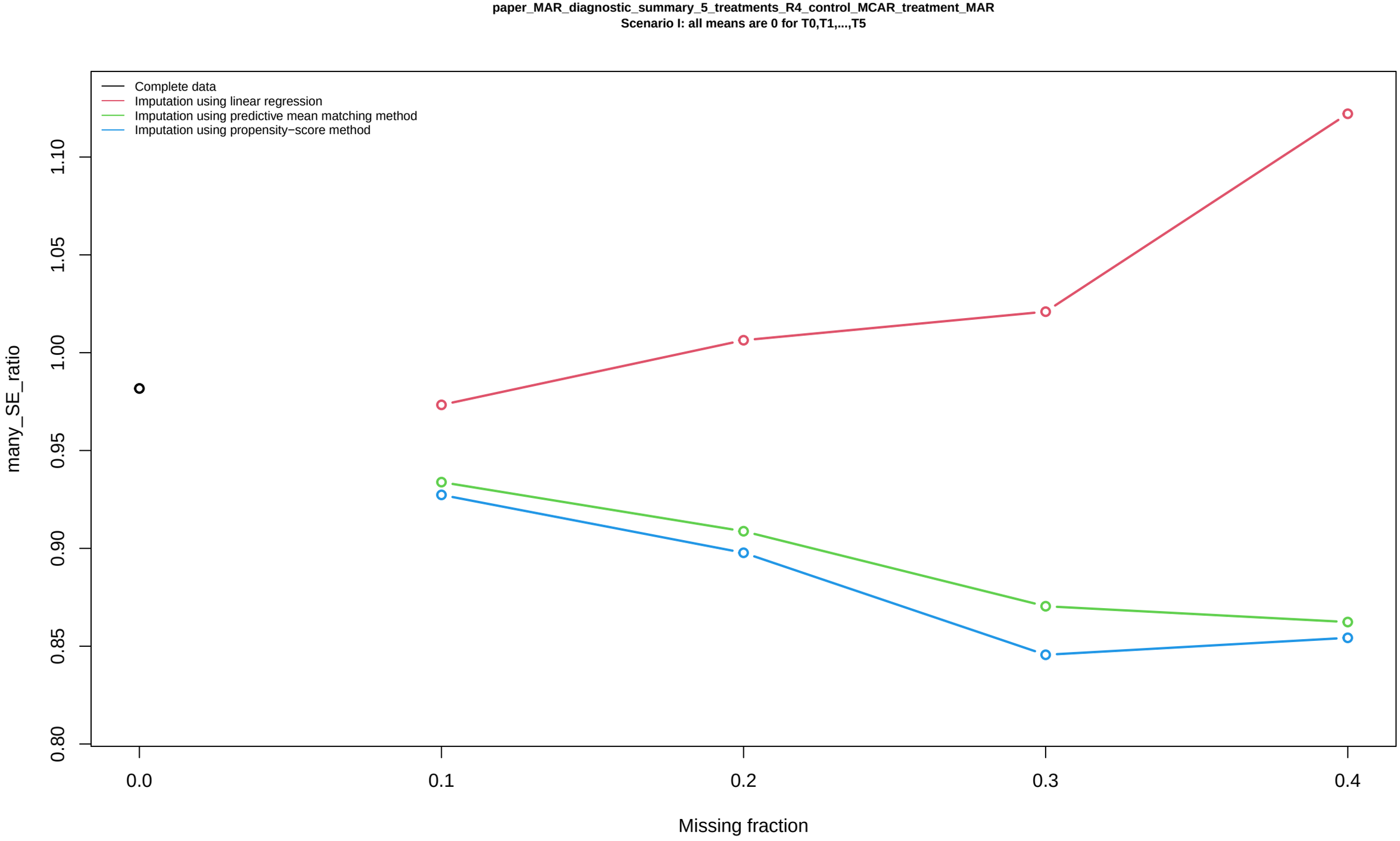

paper_MAR_diagnostic_summary_5_treatments_R2_opposite_sign_MAR
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
bias
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.2
0.0
0.1
0.2
0.3
0.4
Missing fraction

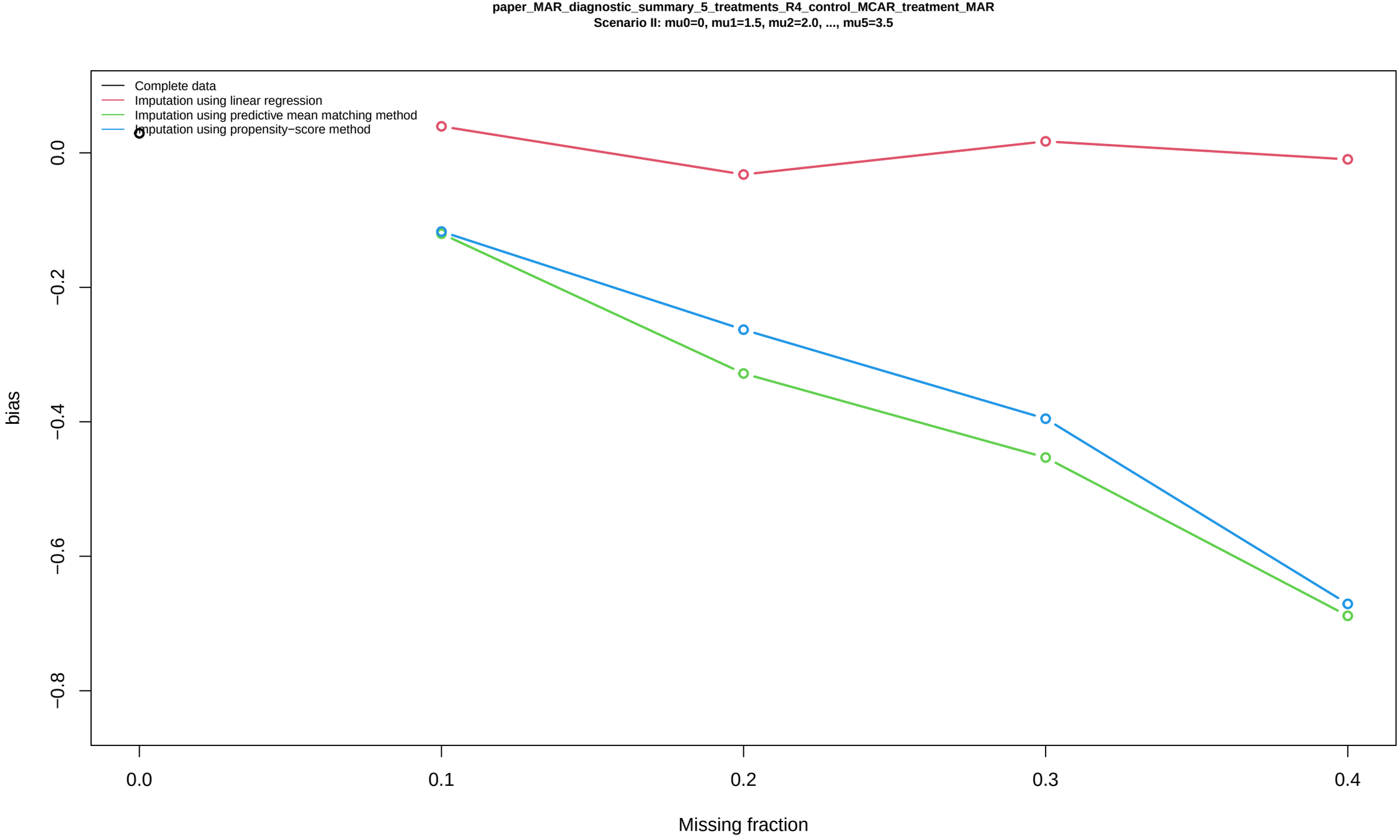

paper_MAR_diagnostic_summary_5_treatments_R2_opposite_sign_MAR
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
RMSE
1.2
1.4
1.6
1.8
2.0
2.2
0.0
0.1
0.2
0.3
0.4
Missing fraction

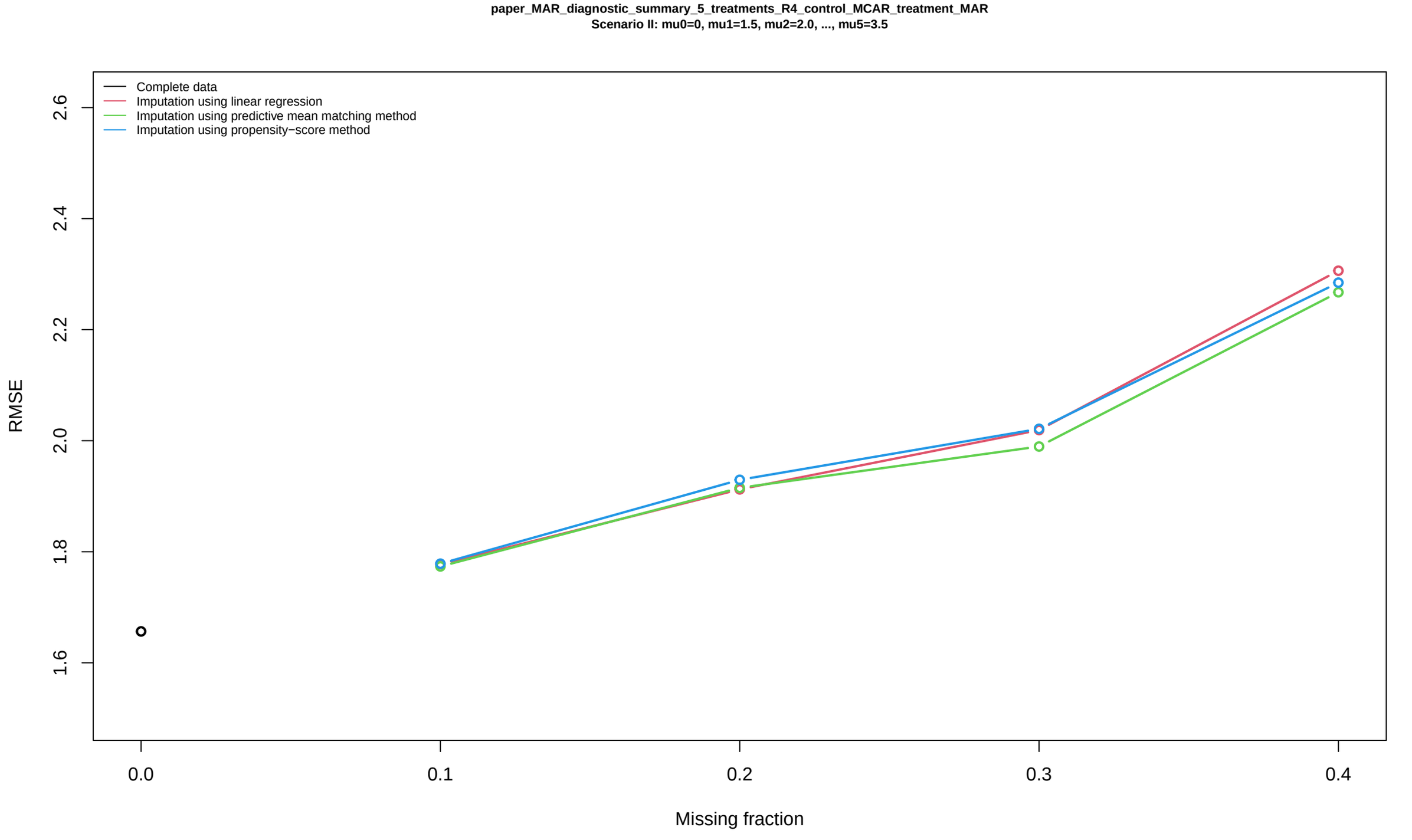

paper_MAR_diagnostic_summary_5_treatments_R2_opposite_sign_MAR
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
many_SE_ratio
1.1
1.0
0.9
0.8
0.0
0.1
0.2
0.3
0.4
Missing fraction

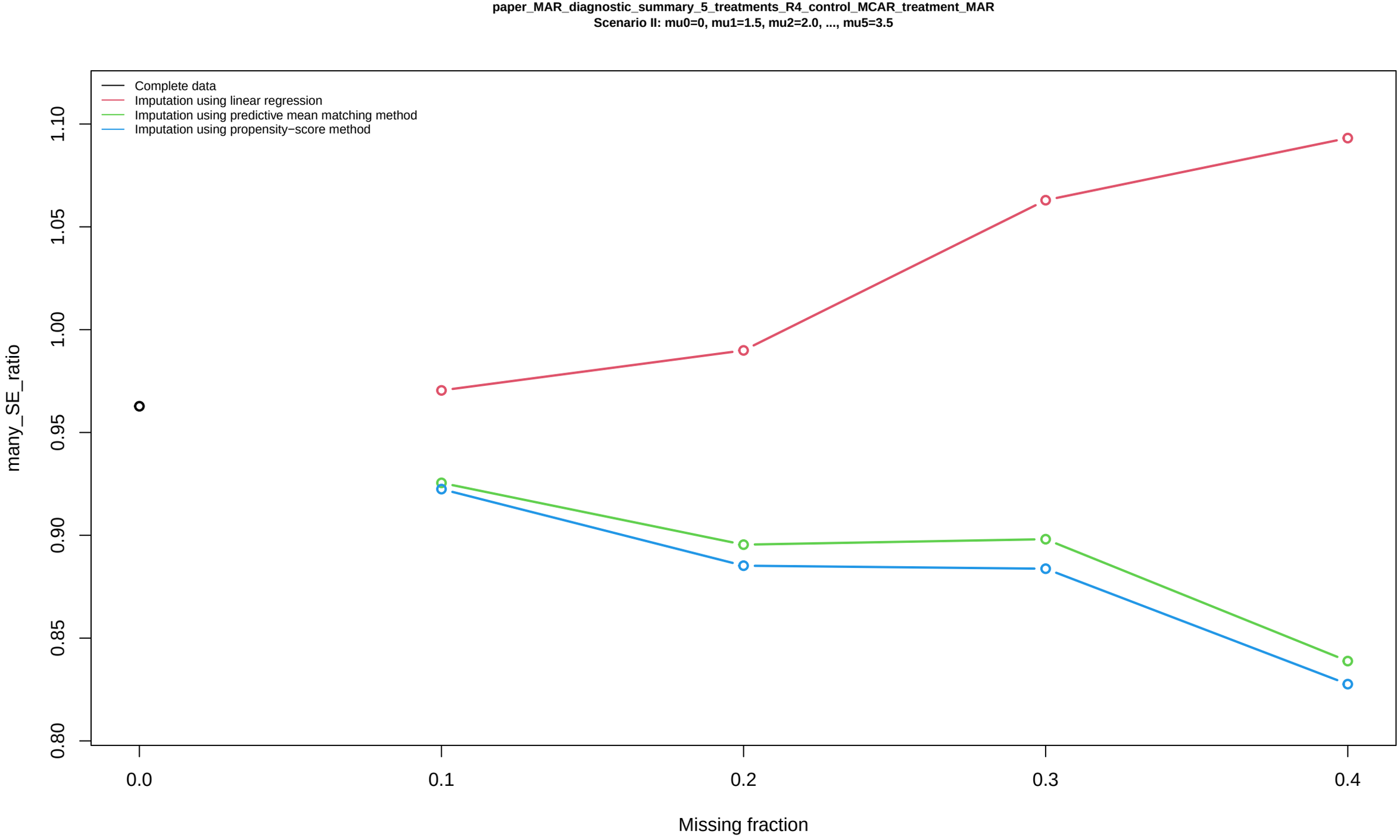

paper_MAR_diagnostic_summary_5_treatments_R4_control_MCAR_treatment_MAR
Scenario I: all means are 0 for T0,T1,...,T5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
bias
0.0
−0.2
−0.4
−0.6
−0.8
0.0
0.1
0.2
0.3
0.4
Missing fraction

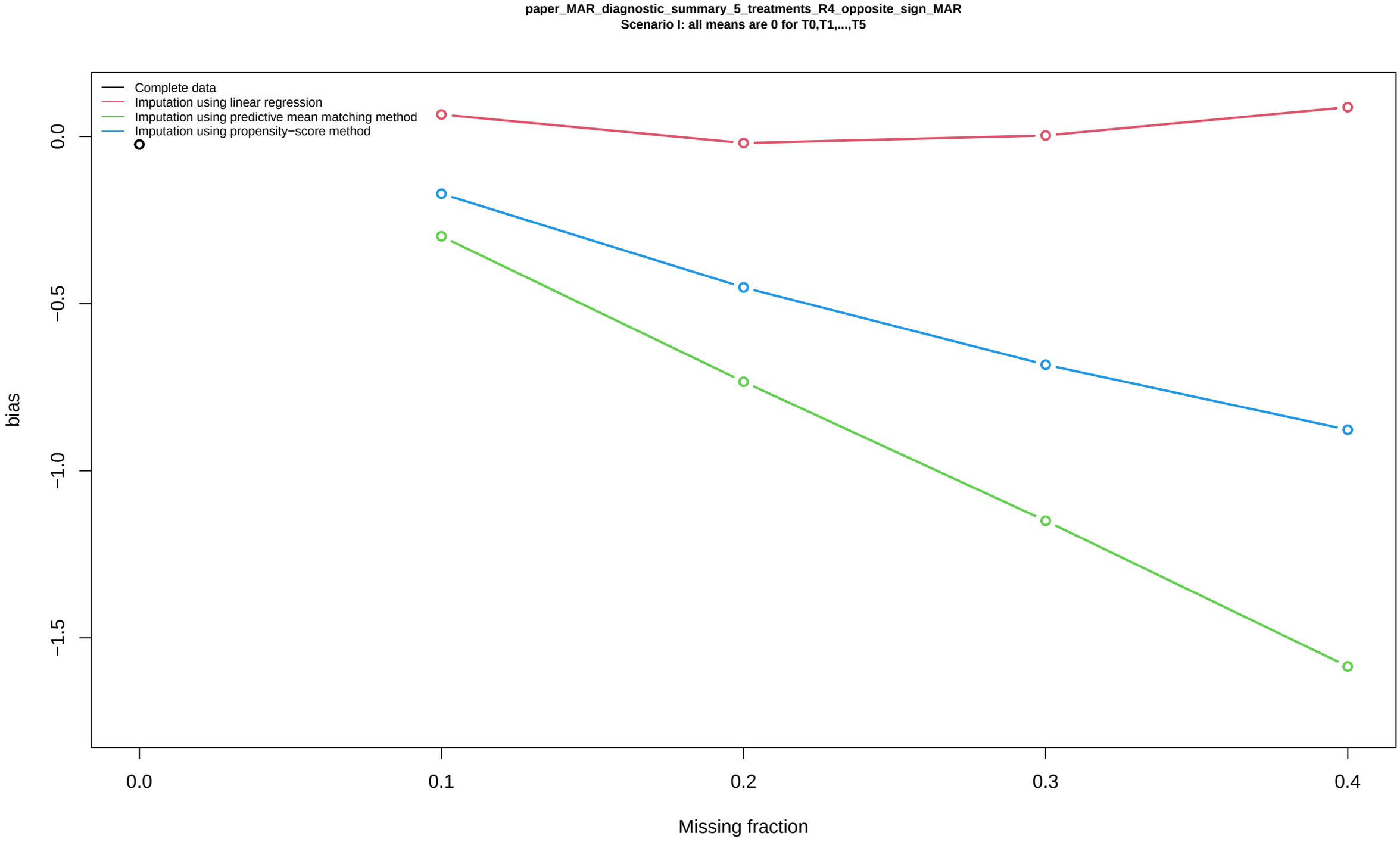

paper_MAR_diagnostic_summary_5_treatments_R4_control_MCAR_treatment_MAR
Scenario I: all means are 0 for T0,T1,...,T5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
RMSE
1.6
1.8
2.0
2.2
2.4
0.0
0.1
0.2
0.3
0.4
Missing fraction

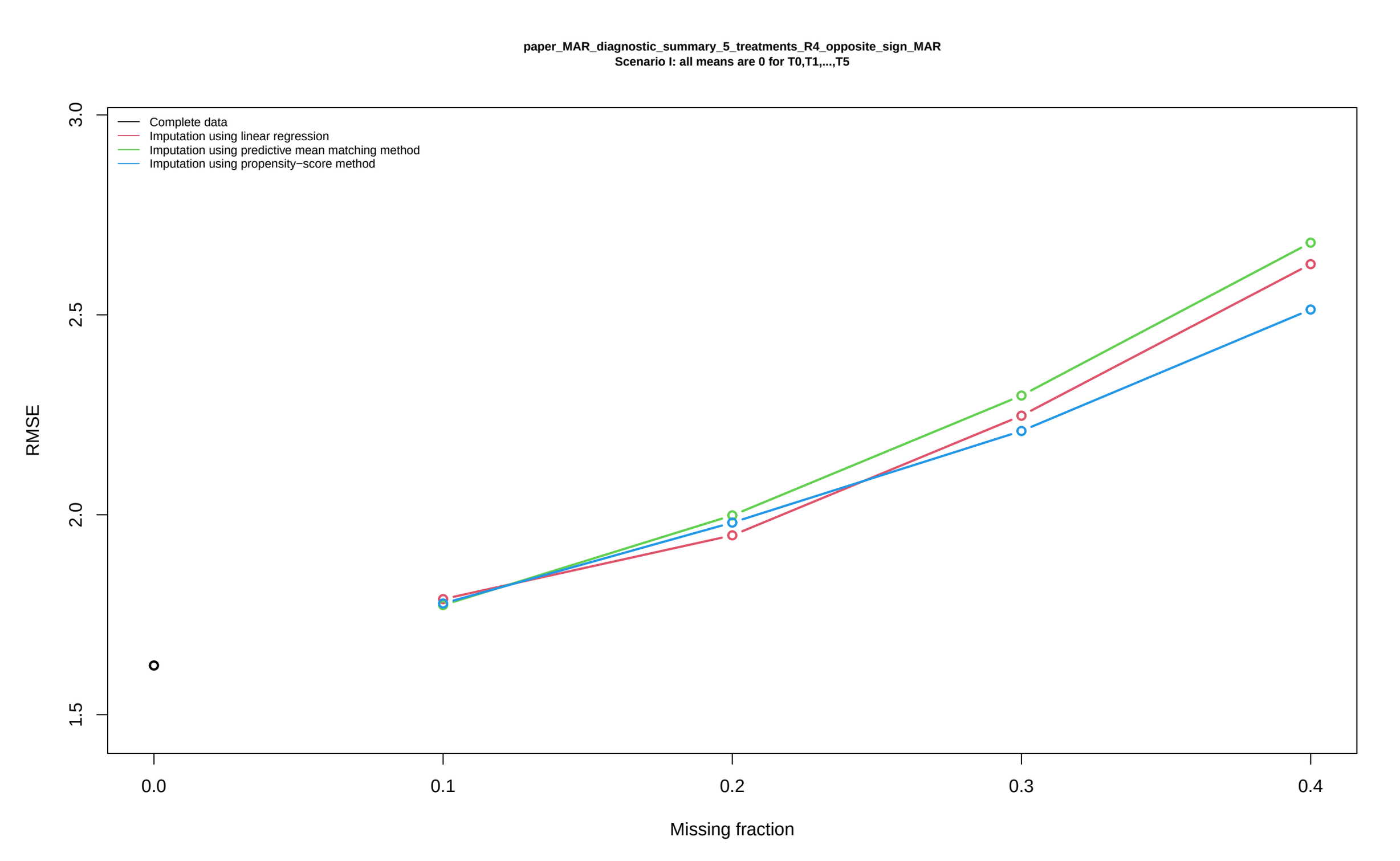

paper_MAR_diagnostic_summary_5_treatments_R4_control_MCAR_treatment_MAR
Scenario I: all means are 0 for T0,T1,...,T5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
many_SE_ratio
0.80
0.85
0.90
0.95
1.00
1.05
1.10
0.0
0.1
0.2
0.3
0.4
Missing fraction

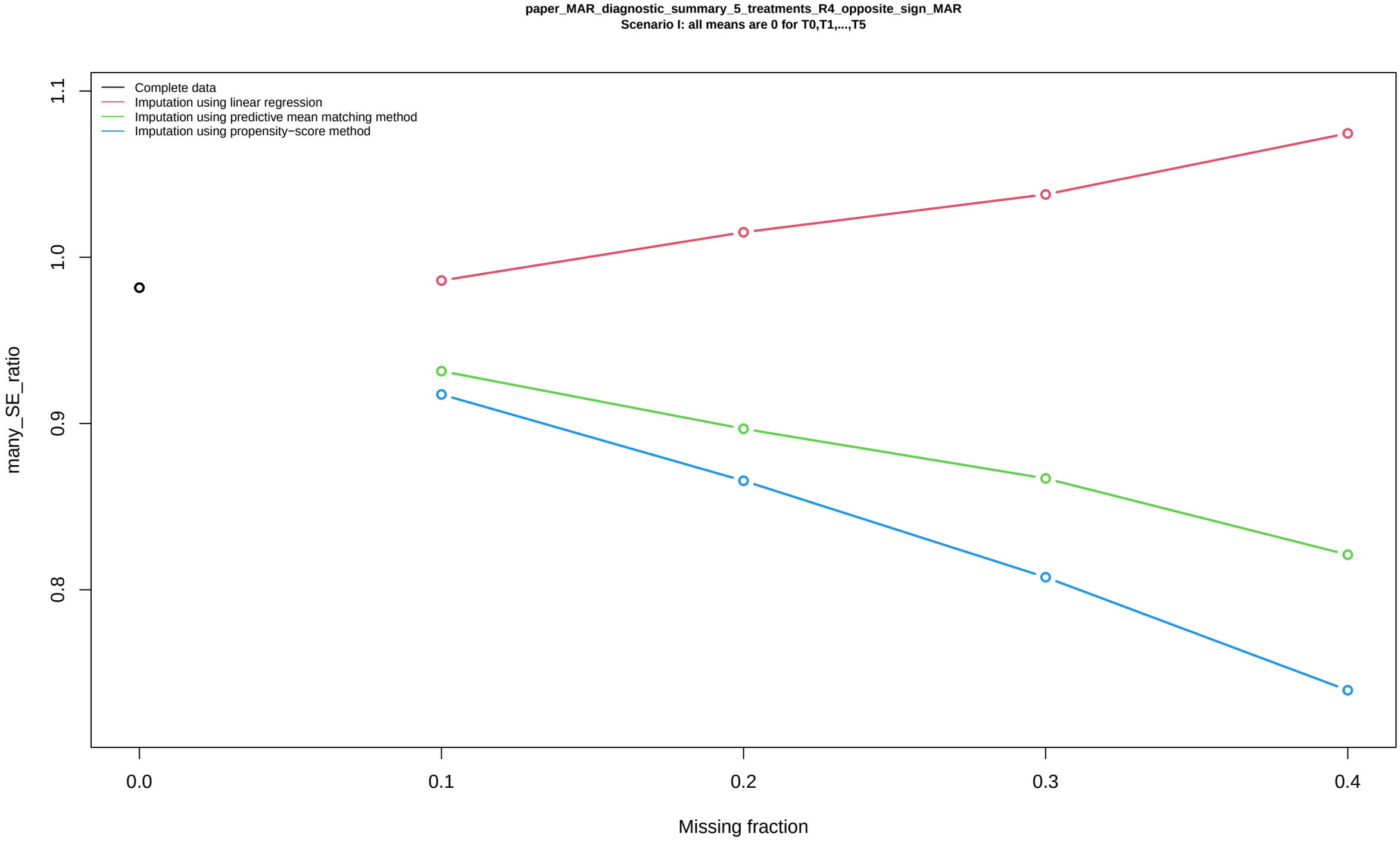

paper_MAR_diagnostic_summary_5_treatments_R4_control_MCAR_treatment_MAR
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
bias
0.0
−0.2
−0.4
−0.6
−0.8
0.0
0.1
0.2
0.3
0.4
Missing fraction

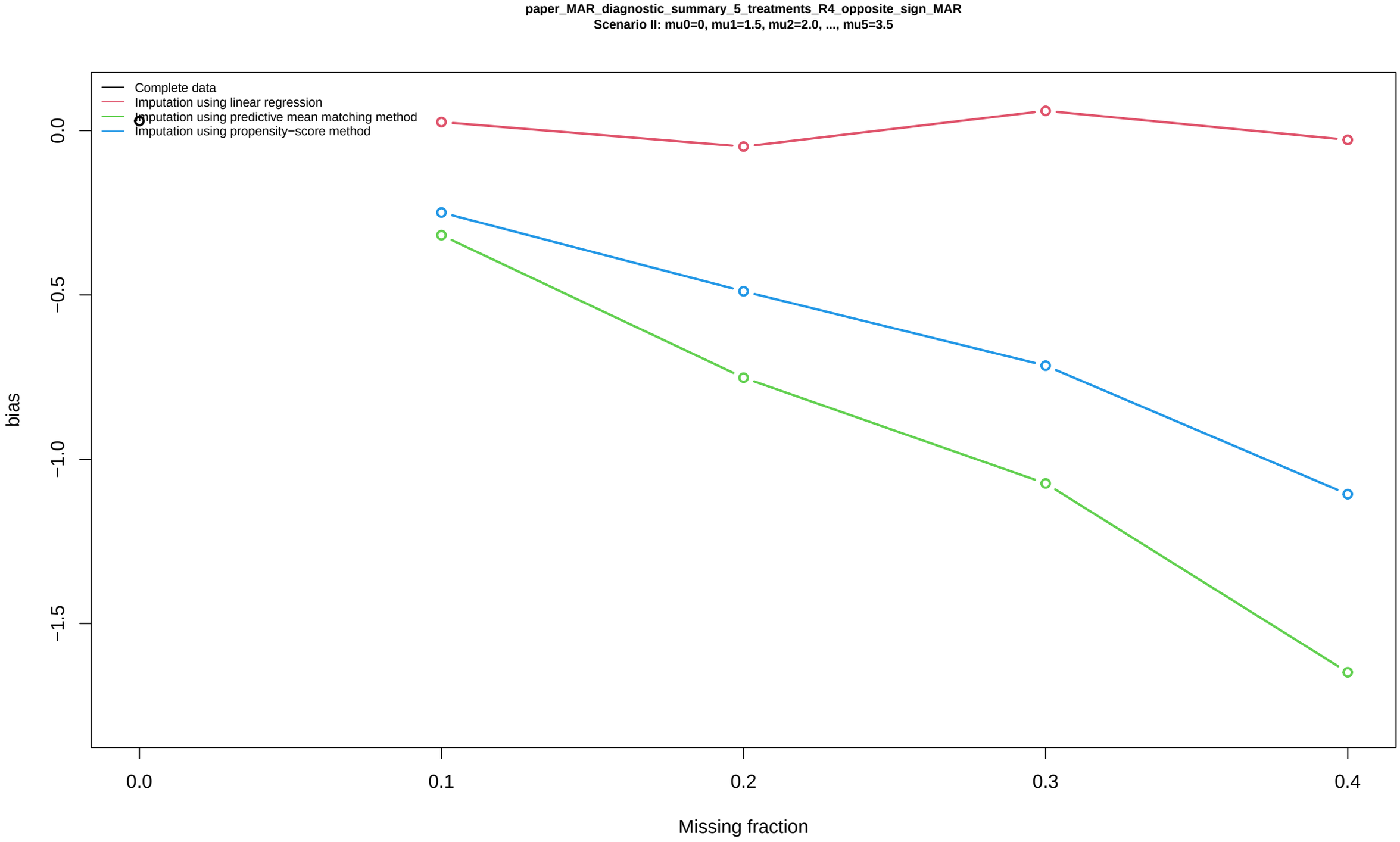

paper_MAR_diagnostic_summary_5_treatments_R4_control_MCAR_treatment_MAR
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
RMSE
1.6
1.8
2.0
2.2
2.4
2.6
0.0
0.1
0.2
0.3
0.4
Missing fraction

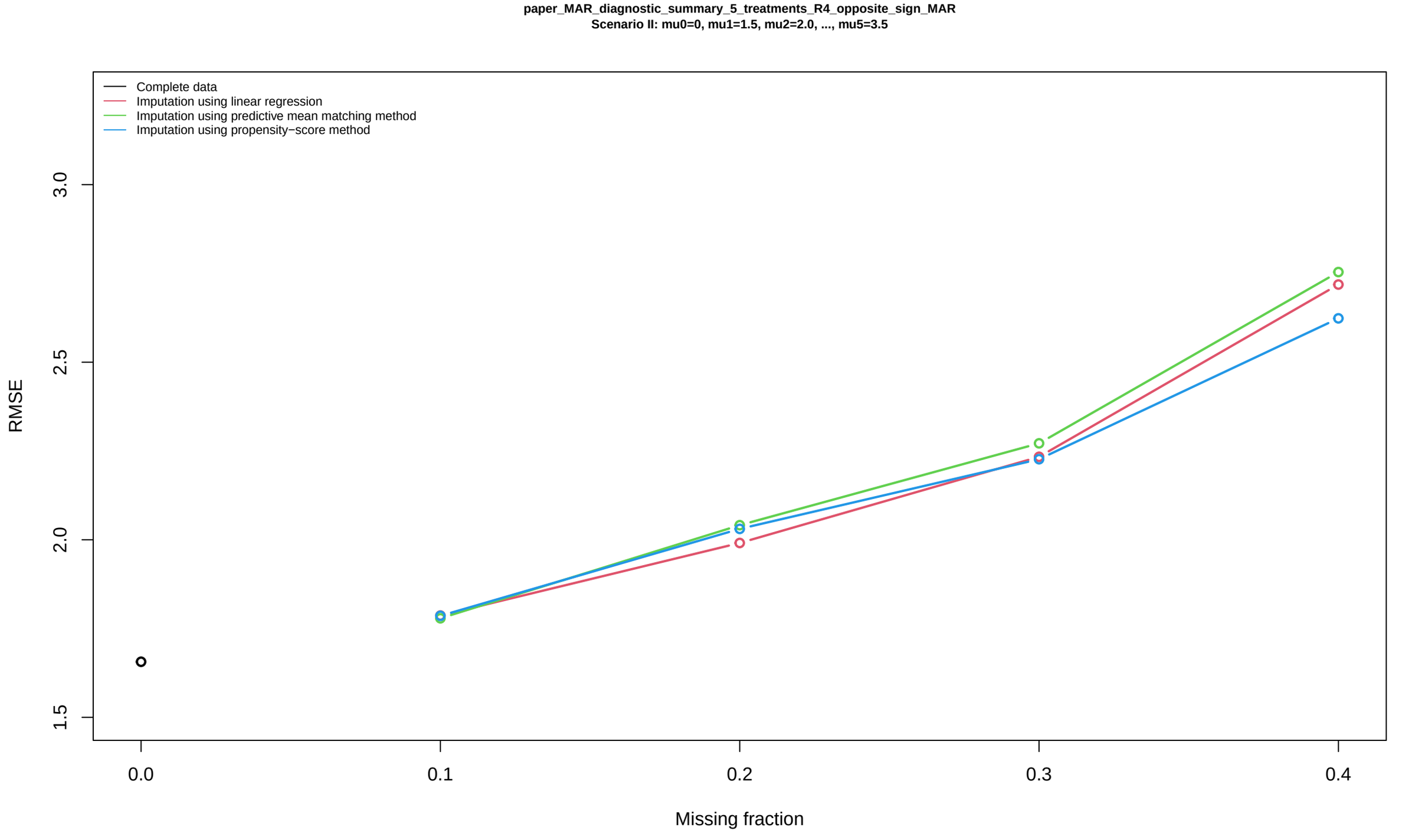

paper_MAR_diagnostic_summary_5_treatments_R4_control_MCAR_treatment_MAR
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
many_SE_ratio
0.80
0.85
0.90
0.95
1.00
1.05
1.10
0.0
0.1
0.2
0.3
0.4
Missing fraction

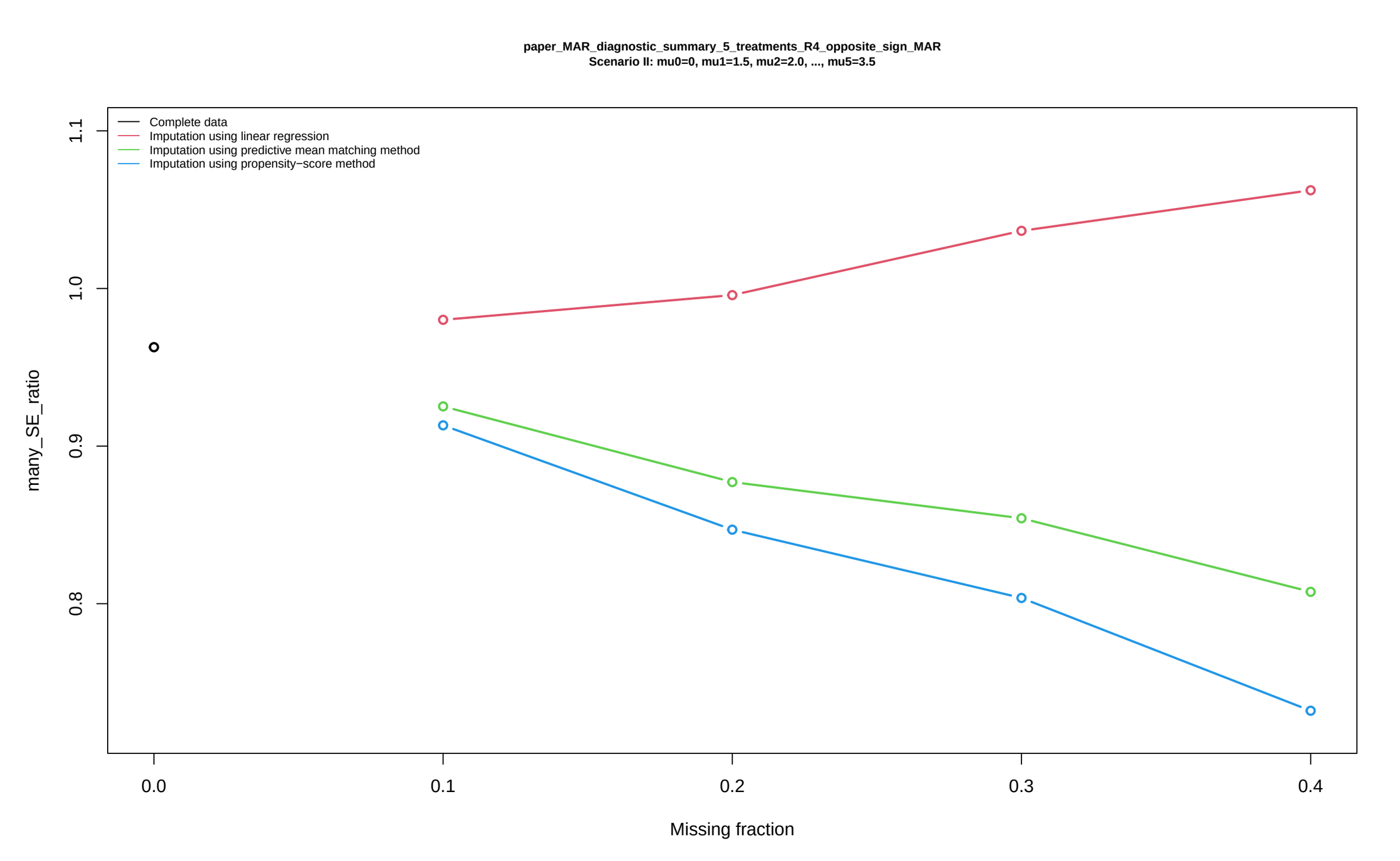

paper_MAR_diagnostic_summary_5_treatments_R4_opposite_sign_MAR
Scenario I: all means are 0 for T0,T1,...,T5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
bias
0.0
−0.5
−1.0
−1.5
0.0
0.1
0.2
0.3
0.4
Missing fraction

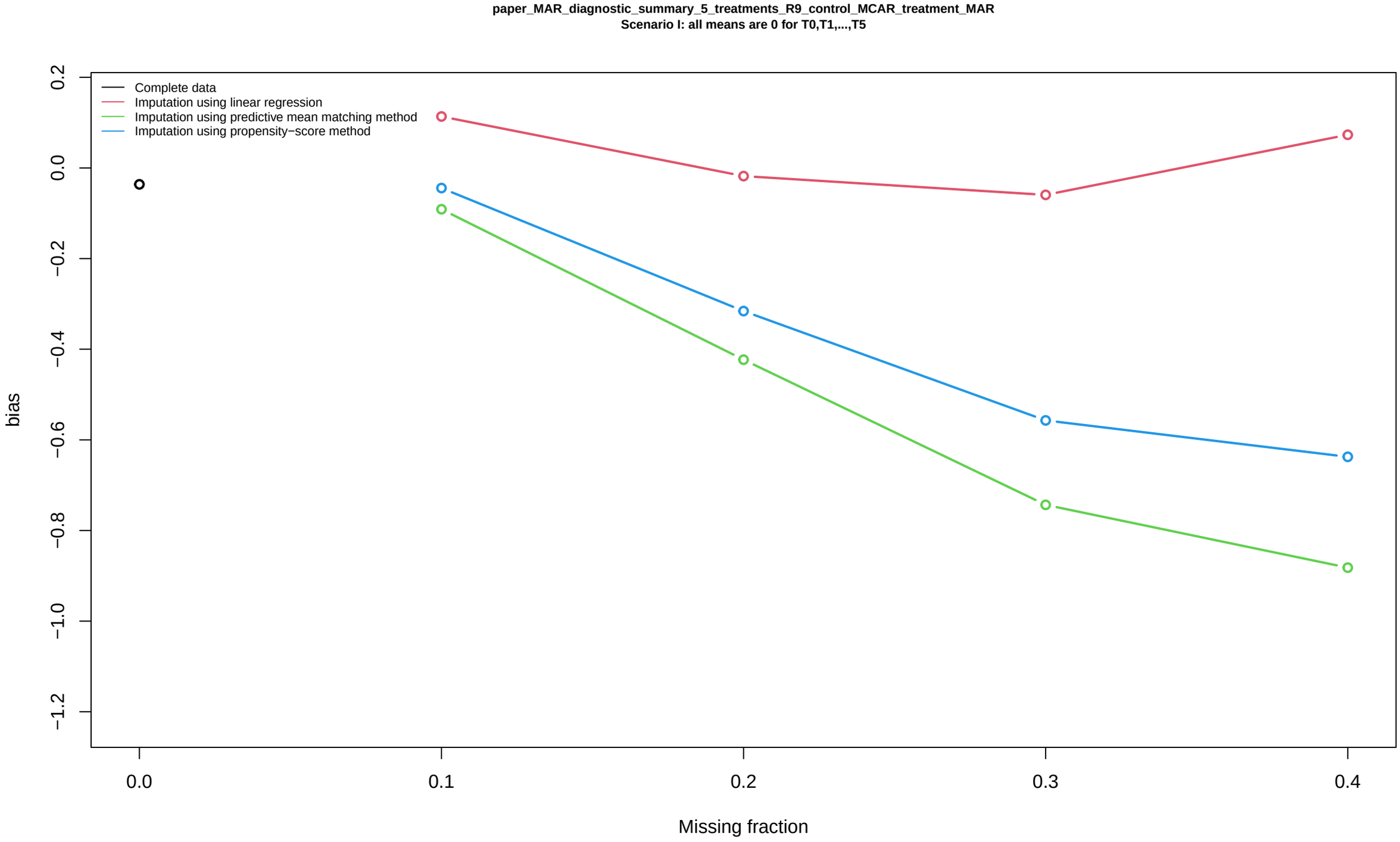

paper_MAR_diagnostic_summary_5_treatments_R4_opposite_sign_MAR
Scenario I: all means are 0 for T0,T1,...,T5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
RMSE
3.0
2.5
2.0
1.5
0.0
0.1
0.2
0.3
0.4
Missing fraction

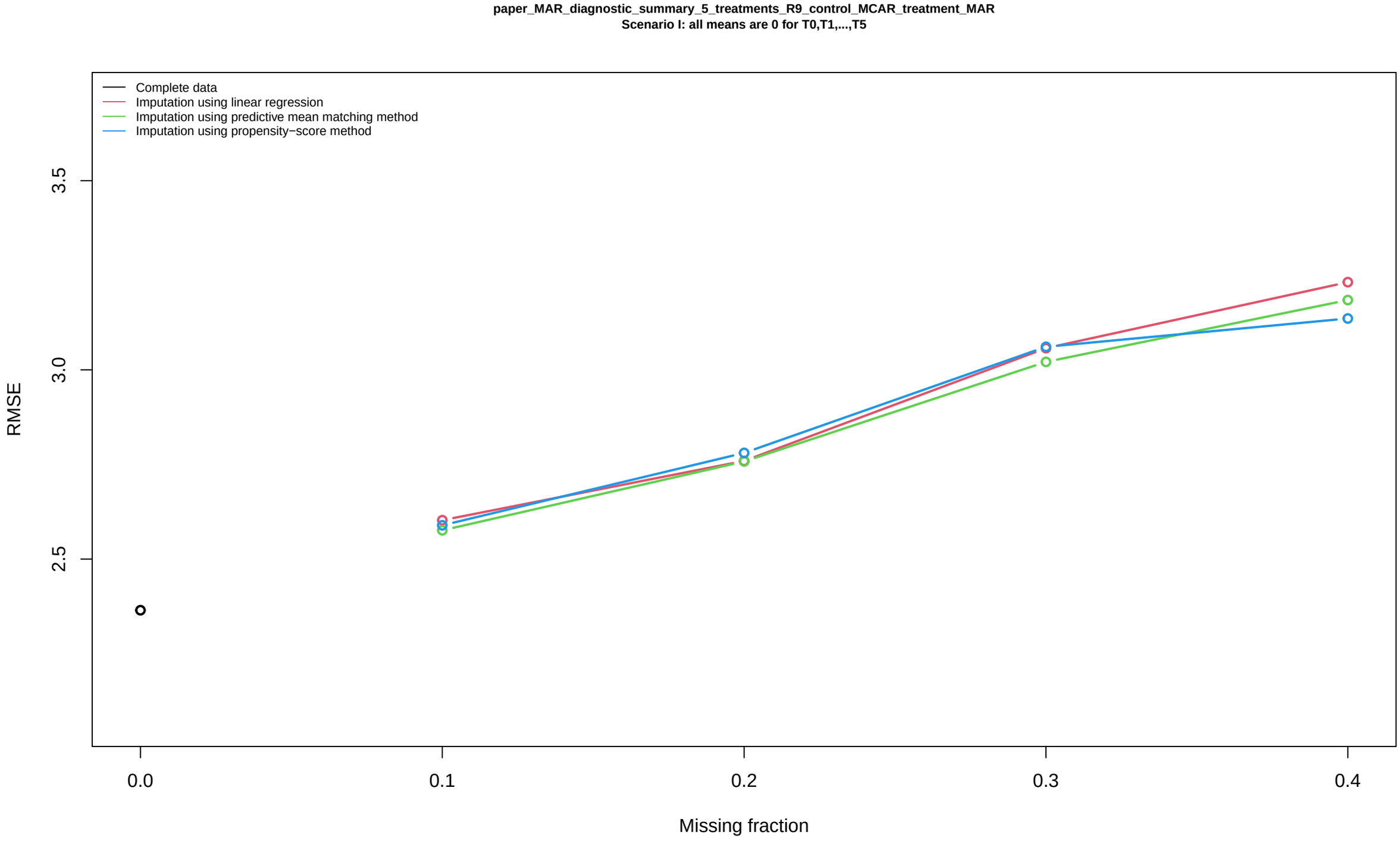

paper_MAR_diagnostic_summary_5_treatments_R4_opposite_sign_MAR
Scenario I: all means are 0 for T0,T1,...,T5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
many_SE_ratio
1.1
1.0
0.9
0.8
0.0
0.1
0.2
0.3
0.4
Missing fraction

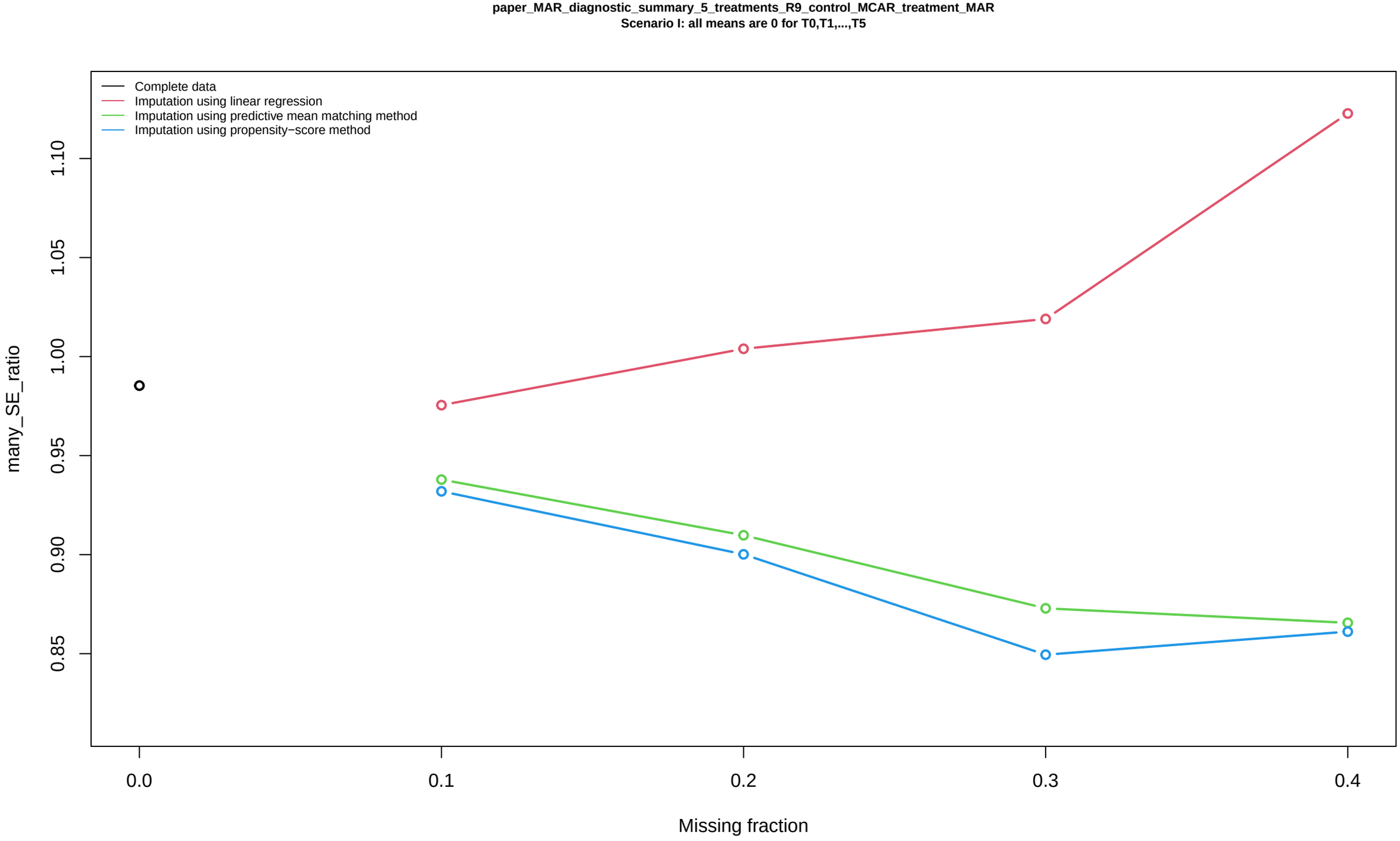

paper_MAR_diagnostic_summary_5_treatments_R4_opposite_sign_MAR
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
bias
0.0
−0.5
−1.0
−1.5
0.0
0.1
0.2
0.3
0.4
Missing fraction

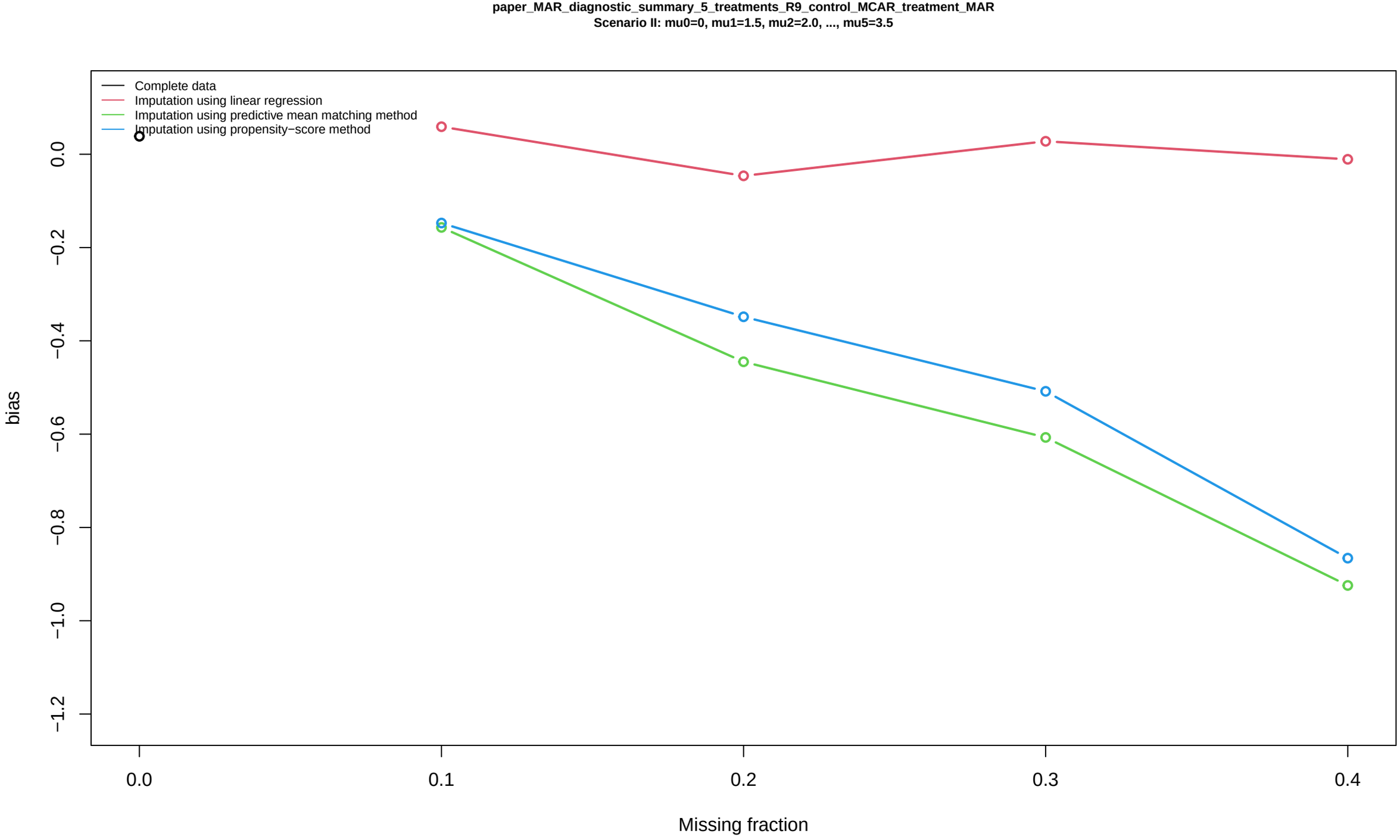

paper_MAR_diagnostic_summary_5_treatments_R4_opposite_sign_MAR
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
RMSE
1.5
2.0
2.5
3.0
0.0
0.1
0.2
0.3
0.4
Missing fraction

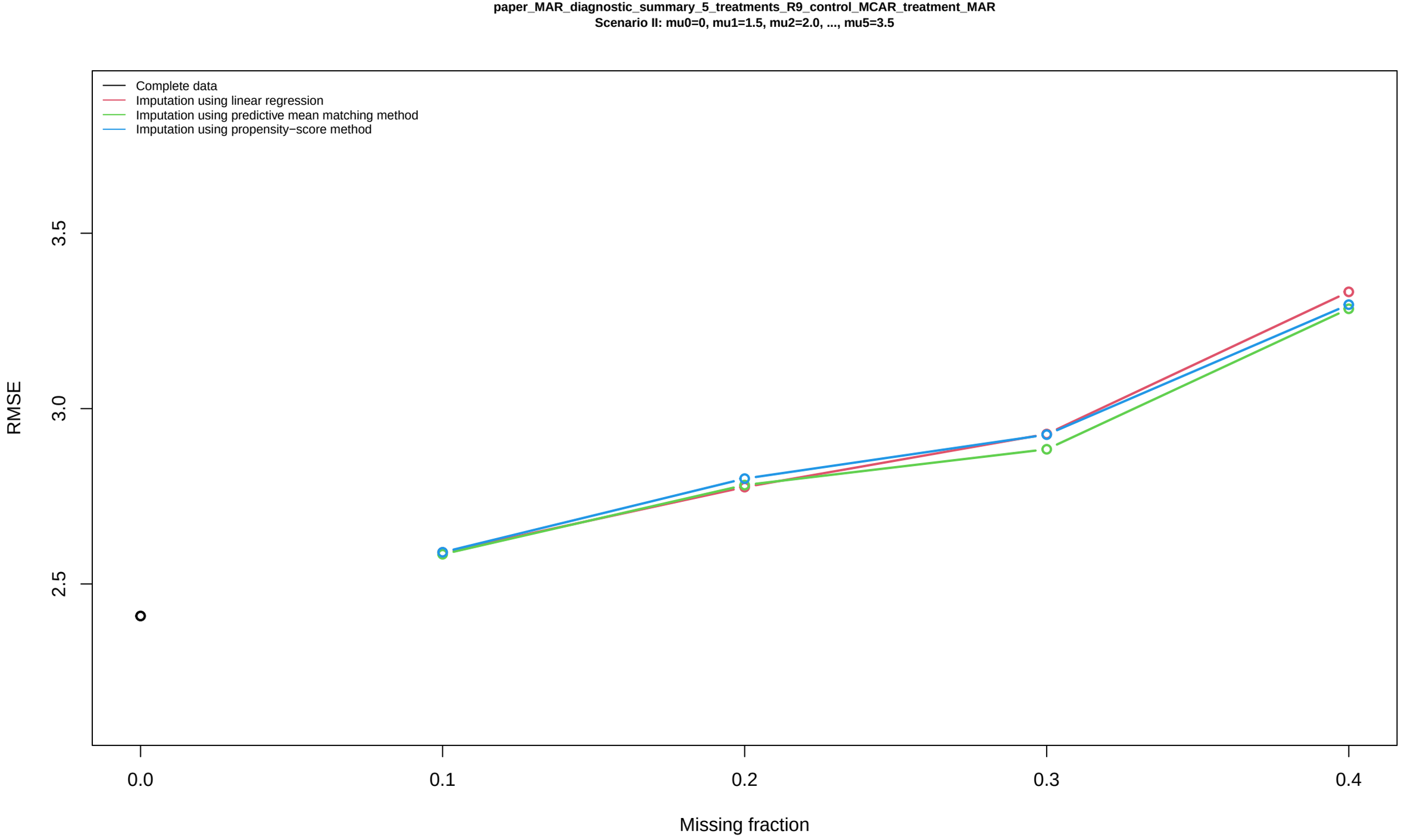

paper_MAR_diagnostic_summary_5_treatments_R4_opposite_sign_MAR
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
many_SE_ratio
1.1
1.0
0.9
0.8
0.0
0.1
0.2
0.3
0.4
Missing fraction

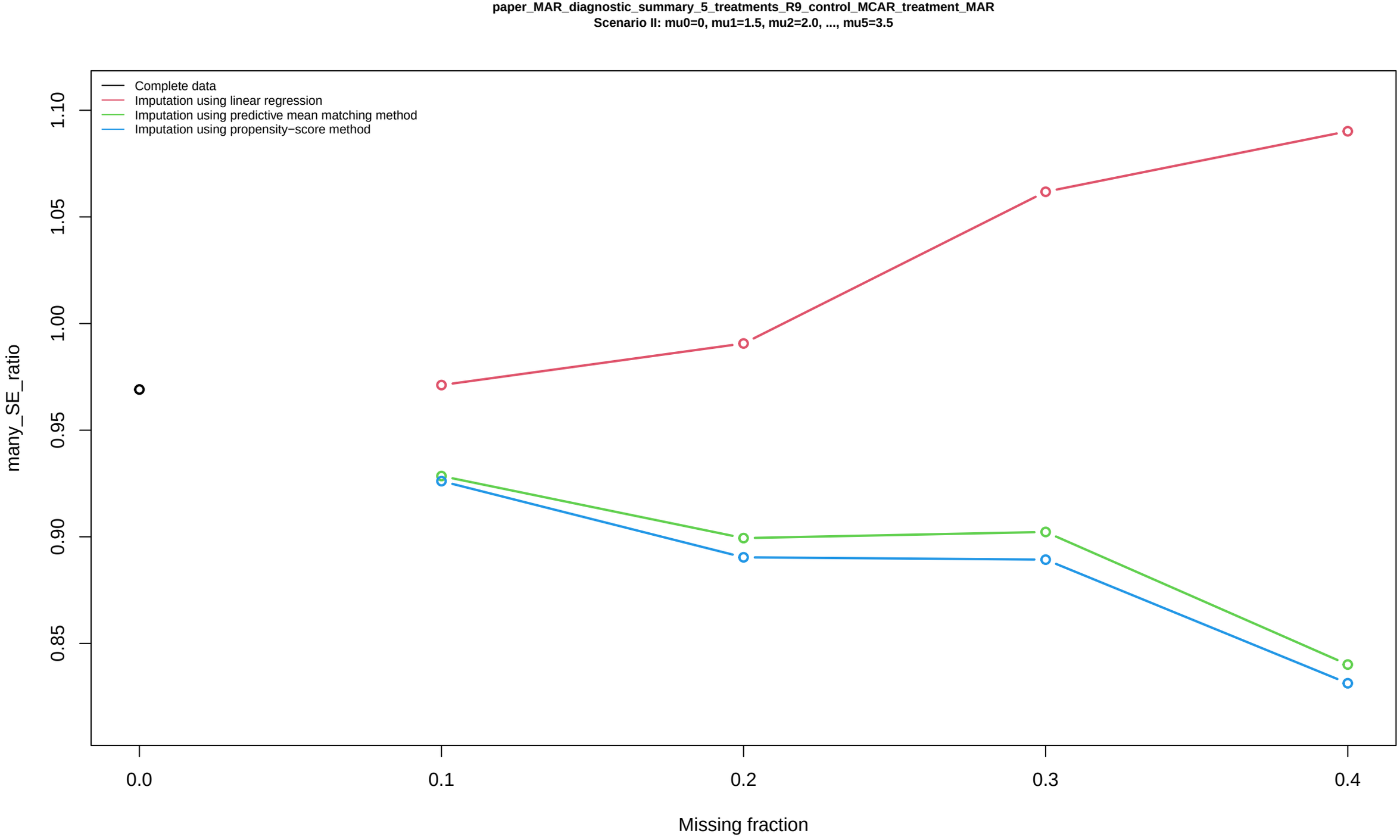

paper_MAR_diagnostic_summary_5_treatments_R9_control_MCAR_treatment_MAR
Scenario I: all means are 0 for T0,T1,...,T5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
bias
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.2
0.0
0.1
0.2
0.3
0.4
Missing fraction

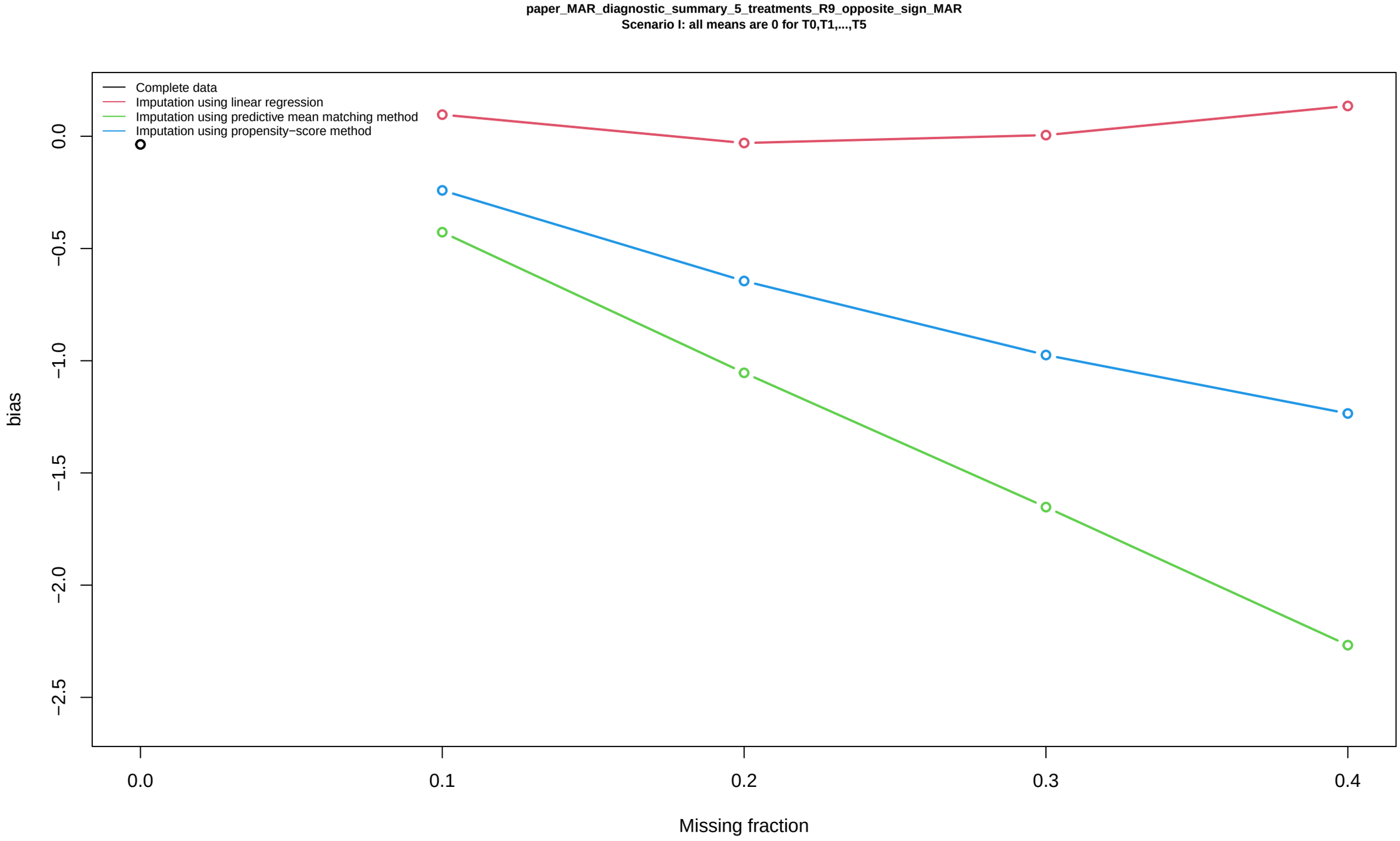

paper_MAR_diagnostic_summary_5_treatments_R9_control_MCAR_treatment_MAR
Scenario I: all means are 0 for T0,T1,...,T5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
RMSE
2.5
3.0
3.5
0.0
0.1
0.2
0.3
0.4
Missing fraction

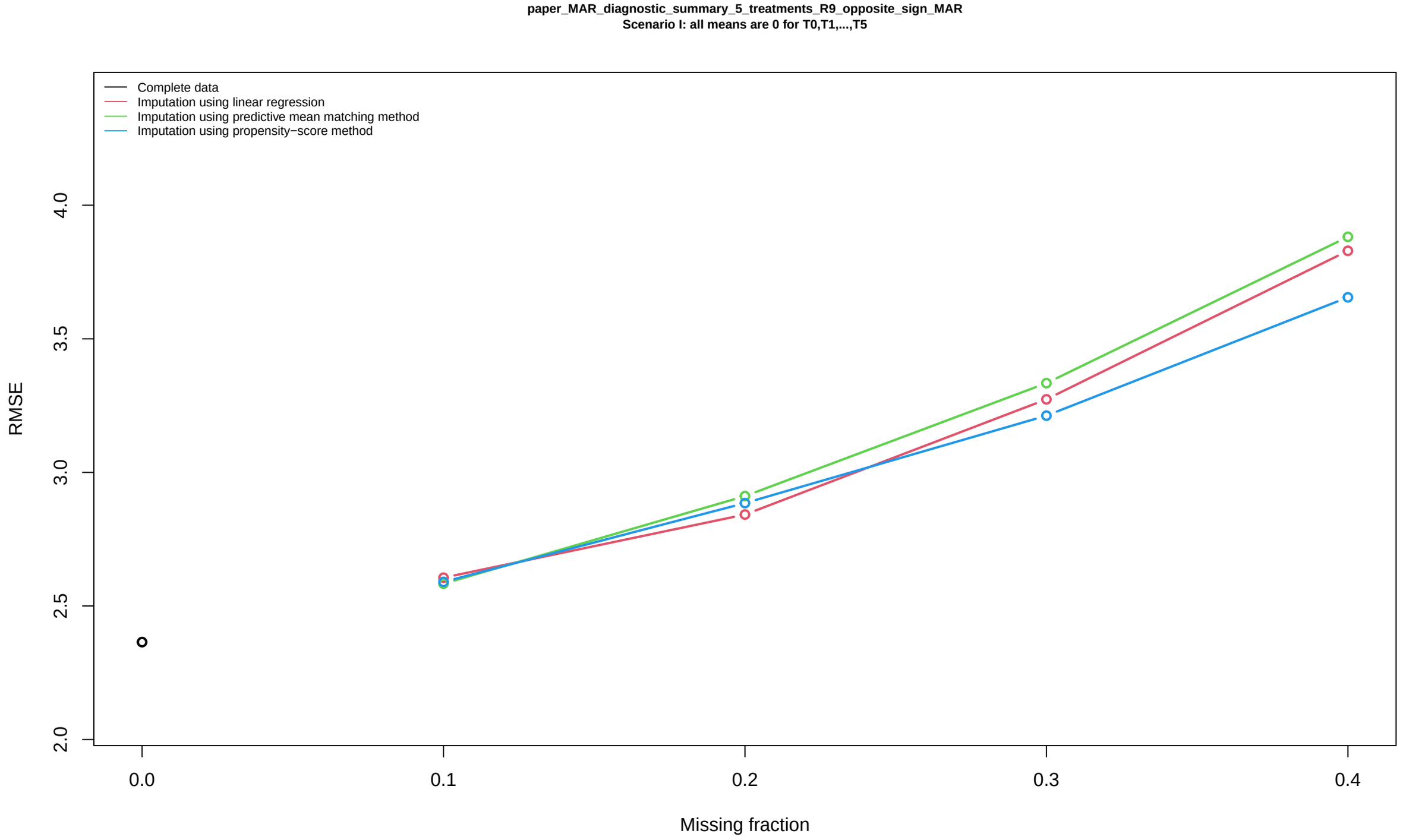

paper_MAR_diagnostic_summary_5_treatments_R9_control_MCAR_treatment_MAR
Scenario I: all means are 0 for T0,T1,...,T5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
many_SE_ratio
0.85
0.90
0.95
1.00
1.05
1.10
0.0
0.1
0.2
0.3
0.4
Missing fraction

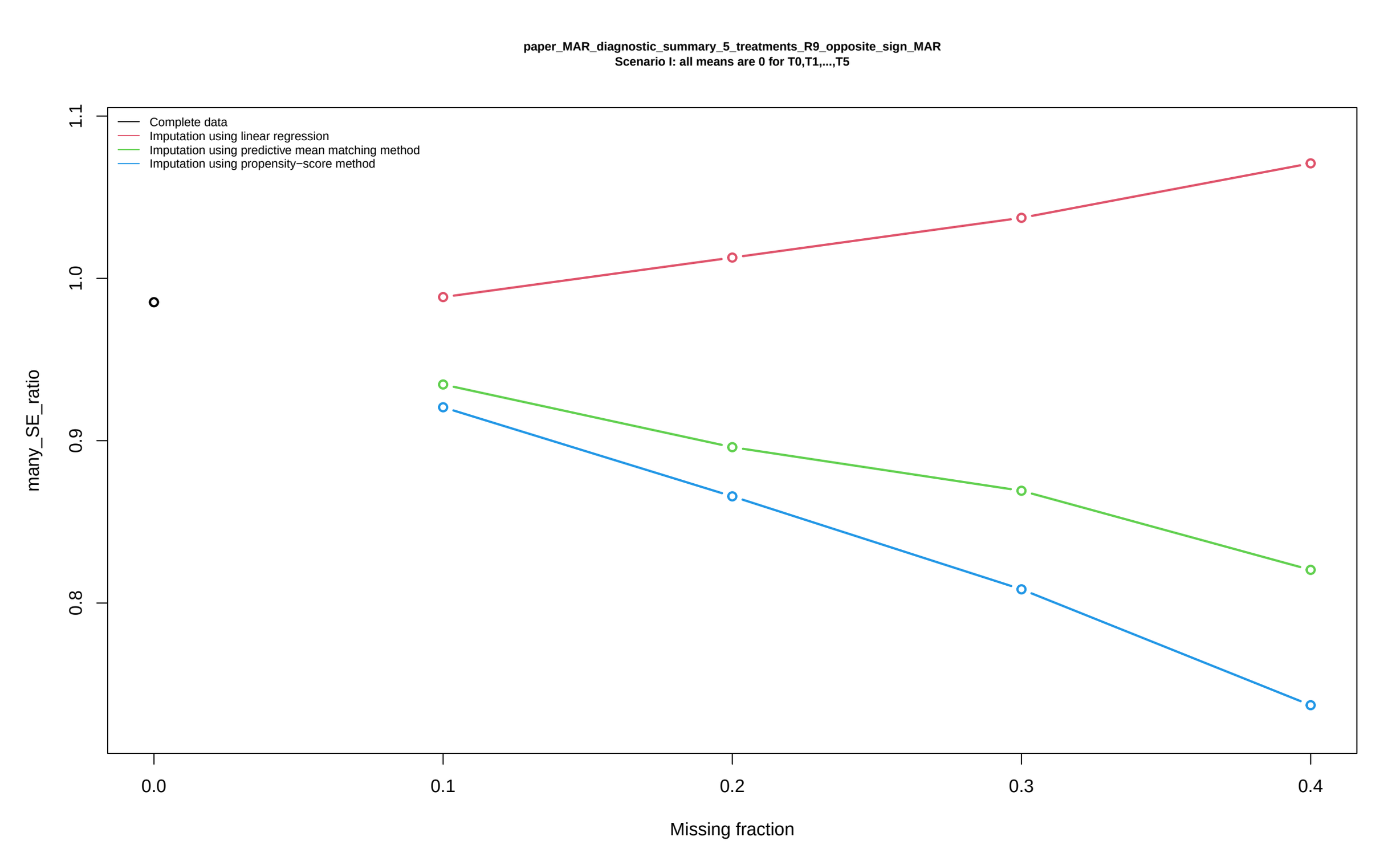

paper_MAR_diagnostic_summary_5_treatments_R9_control_MCAR_treatment_MAR
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
bias
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.2
0.0
0.1
0.2
0.3
0.4
Missing fraction

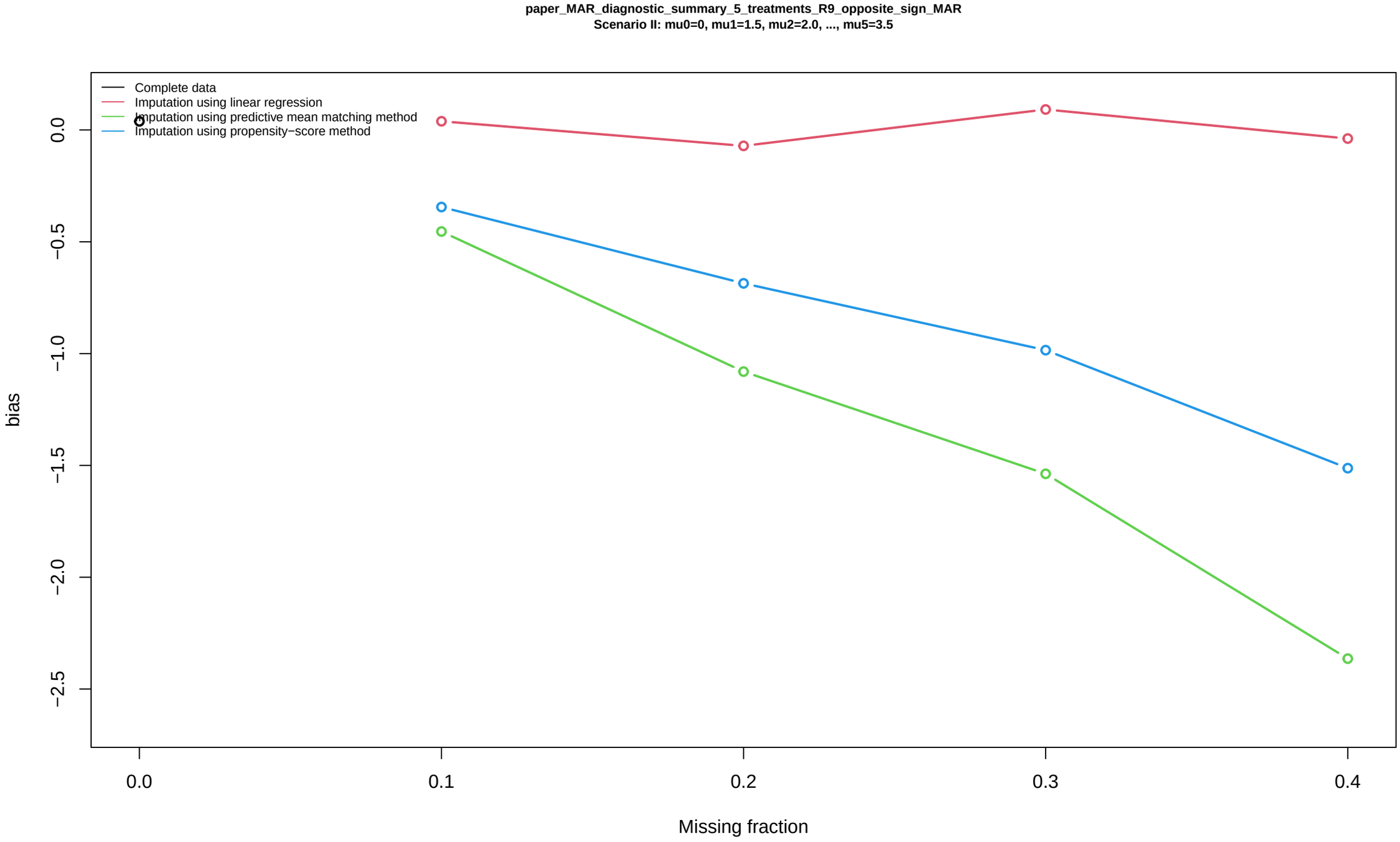

paper_MAR_diagnostic_summary_5_treatments_R9_control_MCAR_treatment_MAR
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
RMSE
2.5
3.0
3.5
0.0
0.1
0.2
0.3
0.4
Missing fraction

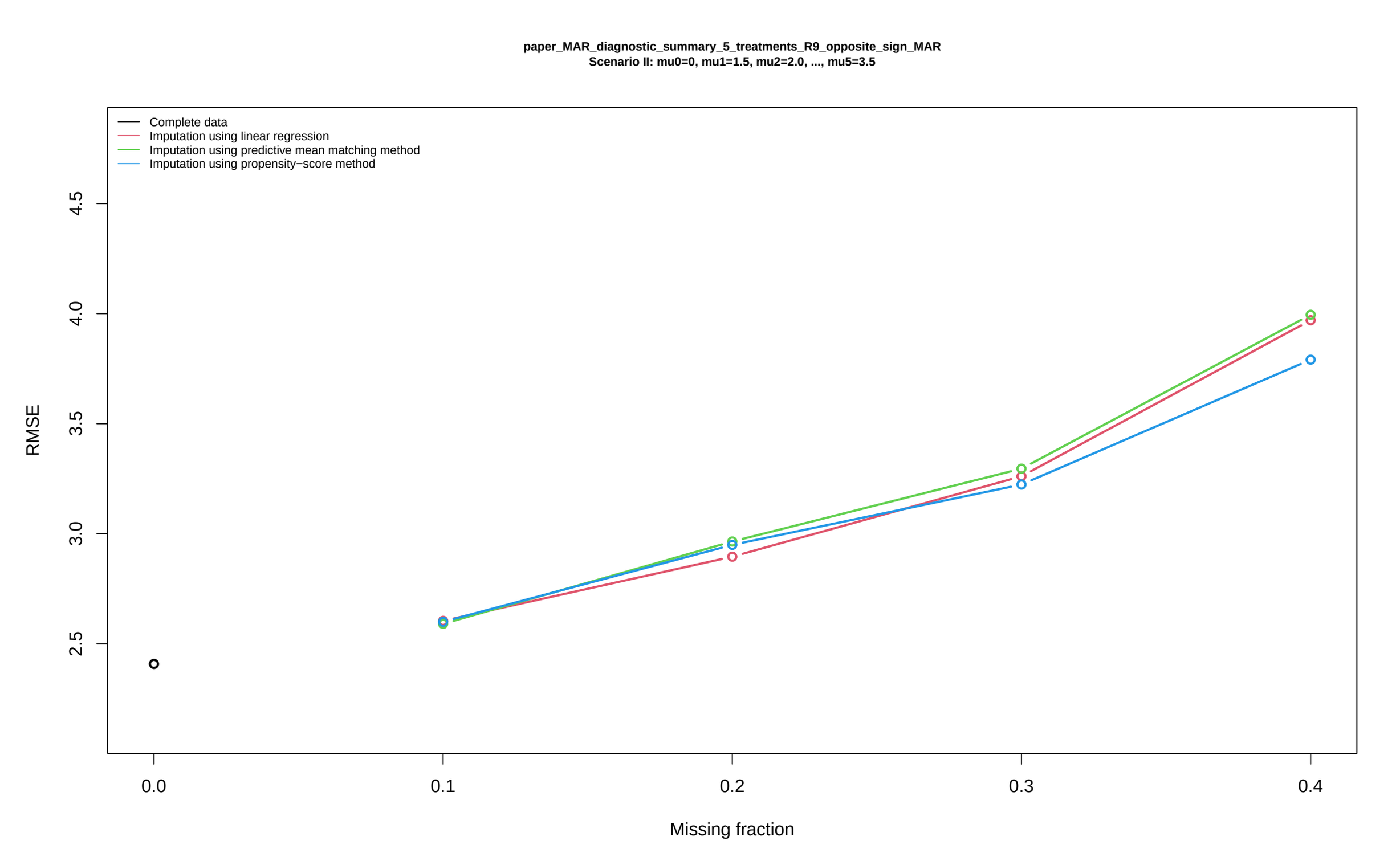

paper_MAR_diagnostic_summary_5_treatments_R9_control_MCAR_treatment_MAR
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
many_SE_ratio
0.85
0.90
0.95
1.00
1.05
1.10
0.0
0.1
0.2
0.3
0.4
Missing fraction

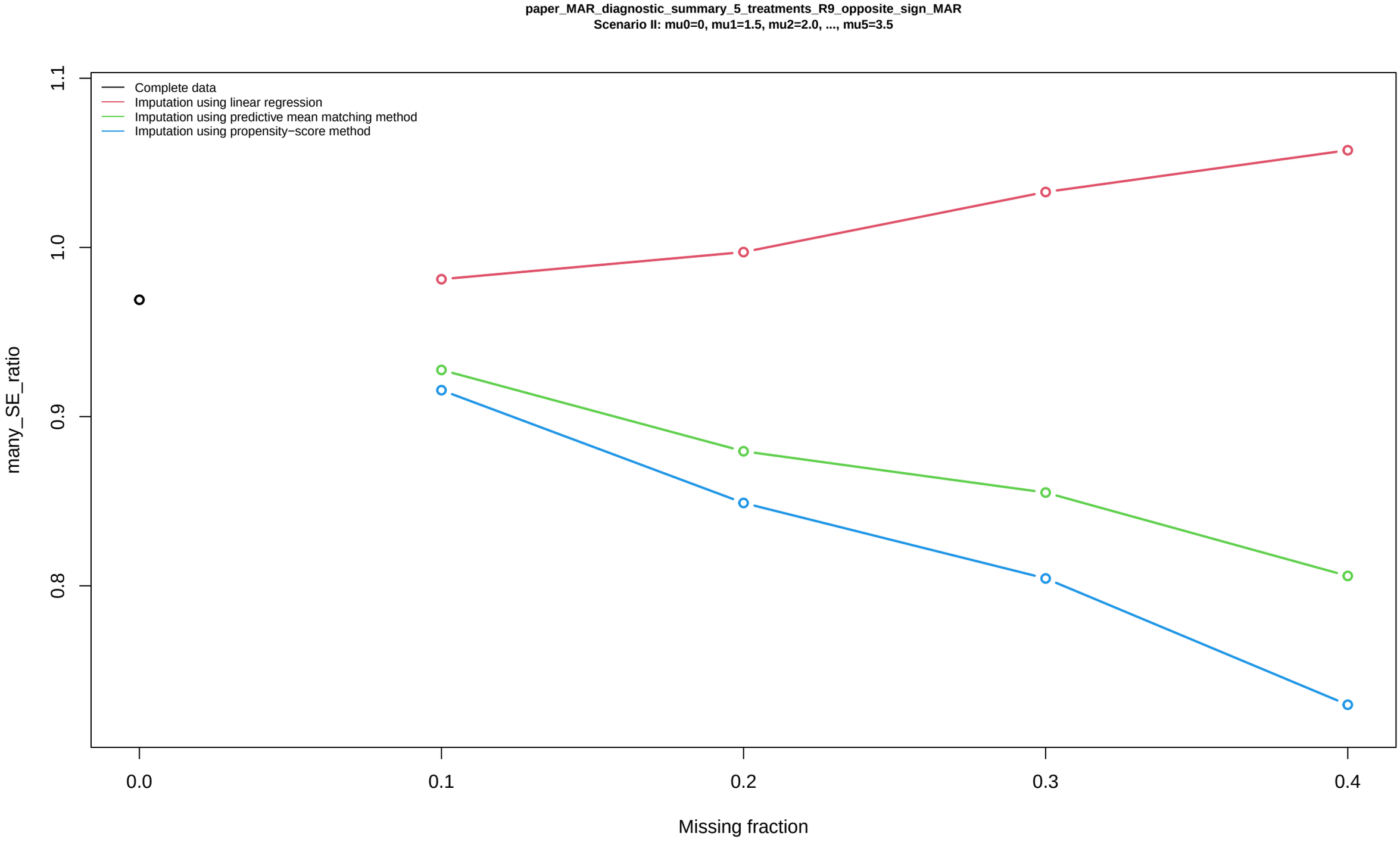

paper_MAR_diagnostic_summary_5_treatments_R9_opposite_sign_MAR
Scenario I: all means are 0 for T0,T1,...,T5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
bias
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
0.0
0.1
0.2
0.3
0.4
Missing fraction


paper_MAR_diagnostic_summary_5_treatments_R9_opposite_sign_MAR
Scenario I: all means are 0 for T0,T1,...,T5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
RMSE
2.0
2.5
3.0
3.5
4.0
0.0
0.1
0.2
0.3
0.4
Missing fraction


paper_MAR_diagnostic_summary_5_treatments_R9_opposite_sign_MAR
Scenario I: all means are 0 for T0,T1,...,T5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
many_SE_ratio
1.1
1.0
0.9
0.8
0.0
0.1
0.2
0.3
0.4
Missing fraction


paper_MAR_diagnostic_summary_5_treatments_R9_opposite_sign_MAR
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
bias
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
0.0
0.1
0.2
0.3
0.4
Missing fraction


paper_MAR_diagnostic_summary_5_treatments_R9_opposite_sign_MAR
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
RMSE
2.5
3.0
3.5
4.0
4.5
0.0
0.1
0.2
0.3
0.4
Missing fraction


paper_MAR_diagnostic_summary_5_treatments_R9_opposite_sign_MAR
Scenario II: mu0=0, mu1=1.5, mu2=2.0, ..., mu5=3.5
Complete data
Imputation using linear regression
Imputation using predictive mean matching method
Imputation using propensity−score method
many_SE_ratio
1.1
1.0
0.9
0.8
0.0
0.1
0.2
0.3
0.4
Missing fraction